Network Science

Concepts, Examples, and Exercises

Your Name

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Table of contents

1 Course Overview **4**
1.1 Welcome . . . 4
1.2 Required packages . . . 4
1.3 Design principles . . . 4
1.4 Current chapters . . . 4

2 Random Graphs **6**
2.1 Introduction . . . 6
2.2 Learning goals . . . 6
2.3 A beginner-friendly motivation . . . 6
2.4 Why the binomial distribution appears . . . 7
2.5 The Erdős–Rényi models . . . 8
2.5.1 The model $G(N, L)$ . . . 8
2.5.2 The model $G(N, p)$ . . . 9
2.5.3 Comparing $G(N, L)$ and $G(N, p)$ . . . 12
2.6 Average degree . . . 12
2.6.1 A quick numerical example . . . 12
2.6.2 Why average degree matters in random graphs . . . 13
2.6.3 Average degree in $G(N, L)$ . . . 14
2.6.4 Average degree in $G(N, p)$ . . . 14
2.6.5 A second way to understand the formula . . . 15
2.6.6 A quick numerical example in $G(N, p)$ . . . 16
2.6.7 Dense and sparse regimes . . . 16
2.6.8 Comparing the two models . . . 16
2.6.9 Summary . . . 17
2.7 Degree distribution . . . 17
2.7.1 A small example . . . 17
2.7.2 From node degrees to degree counts . . . 18
2.7.3 Plotting the degree distribution . . . 19
2.7.4 Degree distribution in the random graph model $G(N, p)$ . . . 20
2.7.5 Interpreting the formula . . . 20
2.7.6 Comparing one realization with the theoretical distribution . . . 20
2.8 Poisson approximation in sparse large graphs . . . 23
2.8.1 Why this approximation appears . . . 23
2.8.2 Class example: binomial distribution and Poisson approximation . . . 23
2.9 Clustering coefficient . . . 24
2.9.1 Local clustering coefficient . . . 24
2.9.2 A fixed node with three different neighborhood structures . . . 25
2.9.3 Example 1 . . . 26
2.9.4 Example 2 . . . 27
2.9.5 Example 3 . . . 27

2.9.6 Summary table for the three examples . . . 27
2.9.7 What clustering is really measuring . . . 28
2.9.8 Average clustering coefficient of a graph . . . 28
2.9.9 Why clustering is important in network science . . . 28
2.9.10 Expected clustering in $G(N, p)$ . . . 28
2.9.11 Why this result matters . . . 29
2.9.12 Numerical check by simulation . . . 29
2.9.13 Summary . . . 31
2.10 Class example: generate and plot a random graph . . . 32
2.11 Interactive explorer . . . 32
2.12 In-class questions . . . 33
2.13 End-of-chapter exercises . . . 33
2.13.1 Part I. Mathematical exercises . . . 33
2.13.2 Part II. Conceptual and analytical exercises . . . 36
2.13.3 Part III. Simulation and coding exercises . . . 37
2.14 Suggested mini-project . . . 39
2.15 Concluding remarks . . . 39

3 The Evolution of Random Graphs 40
3.1 Introduction . . . 40
3.2 Learning goals . . . 40
3.3 Connectedness and components . . . 40
3.3.1 Connected graph . . . 40
3.3.2 Connected component . . . 41
3.3.3 Size of a component . . . 41
3.3.4 Isolated vertex . . . 41
3.3.5 Example: component structure . . . 42
3.3.6 Component-size table . . . 42
3.4 Giant component . . . 43
3.4.1 Definition . . . 43
3.4.2 Why the giant component matters . . . 43
3.5 The evolution of a random graph . . . 43
3.5.1 Evolution through snapshots . . . 44
3.6 Regimes of random-graph evolution . . . 44
3.6.1 Subcritical regime . . . 45
3.6.2 Critical regime . . . 45
3.6.3 Supercritical regime . . . 45
3.6.4 Connected regime . . . 45
3.7 Mathematical description of the giant component . . . 46
3.7.1 Interpretation of the self-consistency equation . . . 46
3.7.2 Near-threshold behavior . . . 46
3.7.3 Structure in the supercritical regime . . . 46
3.7.4 Connectedness threshold . . . 47
3.8 Numerical exploration . . . 47
3.8.1 Largest component fraction . . . 48
3.8.2 Number of connected components . . . 49
3.8.3 Fraction of isolated vertices . . . 50
3.8.4 Probability of connectedness . . . 51
3.9 Summary of regimes . . . 52
3.10 In-class questions . . . 52
3.11 End-of-chapter exercises . . . 53
3.11.1 Part I. Mathematical exercises . . . 53
3.11.2 Part II. Conceptual and analytical exercises . . . 55
3.11.3 Part III. Simulation and coding exercises . . . 56

3.12 Suggested project-style extension 58
3.13 Concluding remarks 58

Chapter 1

Course Overview

1.1 Welcome

This Quarto book is designed as a single, well-structured version of a Network Science course.

Each chapter follows the same pattern:

1. **Theory and intuition**
2. **Worked examples in R**
3. **Exercises and coding tasks**

The same source files produce both:

- an HTML course website
- a PDF course handout

1.2 Required packages

Install the packages below once:

```
install.packages(c(
  "igraph", "dplyr", "ggplot2", "tidyr", "purrr", "plotly", "tinytex", "quarto"
))
tinytex::install_tinytex()
```

1.3 Design principles

This template is built around a few teaching priorities:

- explanations should be readable and beginner-friendly
- formulas should be introduced with intuition, not only stated
- code should be clean enough to use directly in class
- plots should be visually clear and consistent
- HTML pages may contain interactivity, while PDF remains clean and stable

1.4 Current chapters

The current book contains these chapters:

1. **Random Graphs**

2. **The Evolution of Random Graphs**

The first chapter introduces random graph ensembles and their basic structural properties. The second chapter studies how global connectivity emerges as a random graph evolves.

Chapter 2

Random Graphs

2.1 Introduction

Random graphs are one of the most important starting points in network science. In ordinary graph theory, the usual starting point is a fixed graph: the nodes and links are already given, and the goal is to analyse the structure that is already there. In random graph theory, the viewpoint changes. Instead of starting from one fixed graph, the starting point is a rule for generating graphs.

This change is conceptually important. It means that the object of interest is not one network, but a whole collection of possible networks that could be produced by the same random mechanism. Some of these graphs are more likely than others, and many questions are no longer about one exact object. Instead, the natural questions become:

- What is the expected number of links?
- What is the expected average degree?
- What does the degree distribution typically look like?
- How likely is it that triangles or large connected pieces appear?

This is why random graphs are such a useful first model. They provide a clean probabilistic baseline and a simple reference point for later network models.

2.2 Learning goals

By the end of this chapter, it should be possible to:

- explain what a graph ensemble is
- distinguish between the models $G(N, L)$ and $G(N, p)$
- compute expected average degree
- derive the degree distribution in $G(N, p)$
- explain why a Poisson approximation appears in sparse large graphs
- define clustering and interpret what it means
- connect the formulas to simulations in `R`

2.3 A beginner-friendly motivation

A simple way to build intuition is to think about a classroom during the first week of a semester.

Suppose there are N students in a class. At the beginning, not everyone knows everyone else. Over the first few days, pairs of students may talk, exchange contact information, or work together in small activities. If a graph is drawn in which:

- each student is a node
- a link appears when two students become acquainted

then the resulting network depends on chance. Some pairs talk, some do not. Some classes become highly connected, while others remain sparse.

Now imagine that this process is repeated many times with different classes of the same size and under roughly similar conditions. The exact same acquaintance network should not be expected every time. But certain patterns may still be stable:

- the average number of acquaintances per student may be similar
- the probability of observing a student with degree k may follow a predictable distribution
- the network may be denser when the chance of interaction is higher

This is the core idea of a random graph model. The claim is not that real social interaction is random in every detail. Rather, randomness is used as a baseline model. If a simple random rule already explains some feature, that is informative. If a real network differs strongly from the random baseline, that is informative too.

> **Main idea**
>
> A random graph is not a single graph. It is a probability model that generates many possible graphs.

2.4 Why the binomial distribution appears

Before defining the Erdős–Rényi models, it helps to recall the binomial distribution, because it appears naturally in this chapter.

If a random variable X counts the number of successes in n independent trials, where each trial succeeds with probability p, then

$$X \sim B(n, p)$$

and

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Its mean is

$$\mathbb{E}[X] = np$$

and its variance is

$$\mathrm{Var}(X) = np(1-p).$$

This matters because in the model $G(N, p)$, every possible edge is treated as an independent trial:

- success = the edge is present
- failure = the edge is absent

As a result, both the number of edges and the degree of a node can be described using binomial ideas.

2.5 The Erdős–Rényi models

There are two classical random graph models that are closely related but conceptually different.

2.5.1 The model $G(N, L)$

In the model $G(N, L)$:

- the number of nodes is fixed and equal to N
- the number of links is fixed and equal to L
- all simple graphs with exactly those values are equally likely

The word *simple* means that self-loops and multiple edges are excluded.

If there are N nodes, then the total number of possible undirected edges is

$$K = \binom{N}{2} = \frac{N(N-1)}{2}.$$

To construct a graph with exactly L links, L edges must be chosen out of these K possible ones. Therefore, the number of distinct graphs in this ensemble is

$$\Omega(N, L) = \binom{K}{L}.$$

Since each such graph is equally likely, the probability of any one graph G is

$$P(G) = \frac{1}{\binom{K}{L}}.$$

2.5.1.1 Intuition for $G(N, L)$

This model keeps the density tightly controlled because the final graph always has exactly L links. The randomness lies only in **where** those links are placed.

So $G(N, L)$ is useful when all graphs with the same size and the same number of links are being compared.

2.5.1.2 Example: the ensemble $G(3, 2)$

Take three labeled nodes: 1, 2, and 3.

For $N = 3$, the total number of possible undirected edges is

$$K = \binom{3}{2} = 3.$$

These three possible edges are

$$(1, 2), \qquad (1, 3), \qquad (2, 3).$$

Now consider the model $G(3, 2)$. Since exactly two edges must be present, the number of possible labeled graphs is

$$\Omega(3, 2) = \binom{3}{2} = 3.$$

So there are exactly three distinct labeled graphs in this ensemble, and each one has probability

$$P(G) = \frac{1}{3}.$$

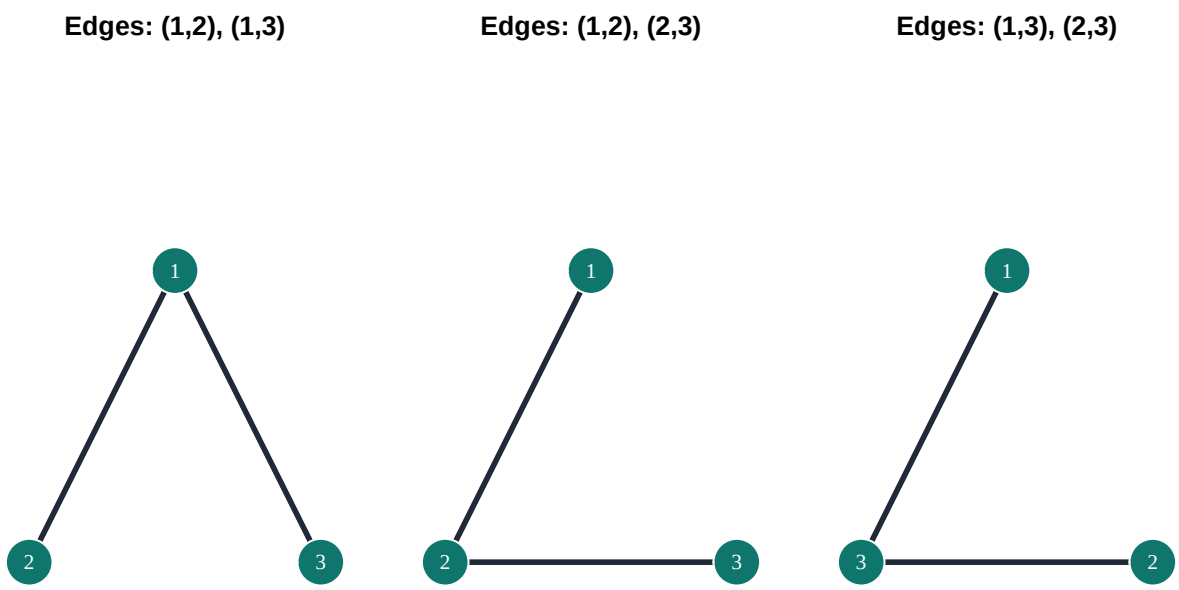


Figure 2.1: The three labeled graphs in the ensemble G(3,2).

This example makes the structure of $G(N, L)$ very concrete:

- the node set is fixed
- the number of edges is fixed
- the only randomness is which L edges are chosen

2.5.2 The model $G(N, p)$

In the model $G(N, p)$:

- the number of nodes is fixed and equal to N
- each possible edge is included independently with probability p

Again, the number of possible undirected edges is

$$K = \binom{N}{2}.$$

Now suppose a generated graph ends up with exactly L links. Then:

- each present edge contributes a factor of p
- each absent edge contributes a factor of $(1 - p)$

Since there are L present edges and $K - L$ absent edges, the probability of that graph is

$$P(G) = p^L (1-p)^{K-L}.$$

2.5.2.1 Intuition for $G(N, p)$

This model does **not** fix the total number of links in advance. Instead, every possible pair of nodes gets its own independent chance to be connected.

That makes $G(N, p)$ especially natural for situations like:

- random failure of communication links
- random activation of pathways
- random acquaintance formation in a large group
- a baseline model where every pair has the same connection probability

2.5.2.2 Example: the ensemble $G(3, p)$

Again take three labeled nodes: 1, 2, and 3.

There are still exactly three possible edges:

$$(1, 2), \qquad (1, 3), \qquad (2, 3).$$

In the model $G(3, p)$, each of these three edges may be either:

- present, with probability p
- absent, with probability $1 - p$

Since each edge has two possibilities and the three edge decisions are independent, the total number of distinct labeled graphs is

$$2^3 = 8.$$

So the ensemble $G(3, p)$ contains eight labeled graphs.

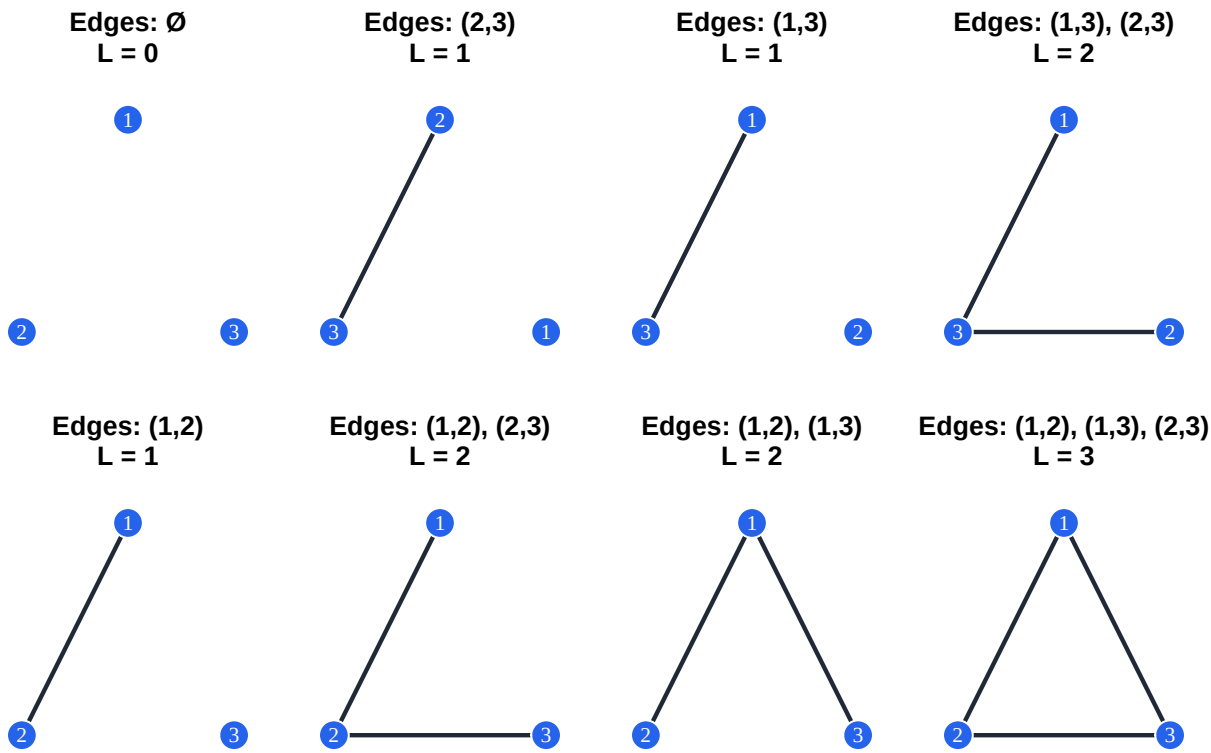


Figure 2.2: The eight labeled graphs in the ensemble G(3,p).

In this case, the number of edges is not fixed. Some of the eight graphs have:

- $L = 0$ edges
- $L = 1$ edge
- $L = 2$ edges
- $L = 3$ edges

What matters is that each possible edge is decided independently.

2.5.2.3 Probability table for $G(3, p)$

Since each graph is determined by the presence or absence of the three possible edges, its probability depends only on how many edges it contains.

Graph	Present edges	Number of edges L	Probability
1	$\emptyset$	0	$(1-p)^3$
2	$(1,2)$	1	$p(1-p)^2$
3	$(1,3)$	1	$p(1-p)^2$
4	$(2,3)$	1	$p(1-p)^2$
5	$(1,2),(1,3)$	2	$p^2(1-p)$
6	$(1,2),(2,3)$	2	$p^2(1-p)$
7	$(1,3),(2,3)$	2	$p^2(1-p)$
8	$(1,2),(1,3),(2,3)$	3	p^3

This shows that in $G(3, p)$ all eight labeled graphs are allowed, but they are not equally likely. Their probabilities depend on how many edges are present.

2.5.3 Comparing $G(N, L)$ and $G(N, p)$

The two models are often confused at first, so it is worth separating them clearly.

- In $G(N, L)$, the number of edges is fixed.
- In $G(N, p)$, the number of edges is random.
- In $G(N, L)$, all allowed graphs are equally likely.
- In $G(N, p)$, graphs with the same number of links have the same probability, but graphs with different numbers of links generally do not.

> **!** Conceptual difference
>
> The model $G(N, L)$ is a uniform choice among graphs with exactly L edges.
> The model $G(N, p)$ is an independent decision process for each possible edge.

2.6 Average degree

Average degree is one of the first global quantities used to summarize a network. It gives a single number that describes, on average, how many neighbors each node has.

If a graph has N nodes and L links, then the average degree is

$$\bar{k} = \frac{1}{N} \sum_{i=1}^{N} k_i,$$

where k_i is the degree of node i.

Using the handshake identity,

$$\sum_{i=1}^{N} k_i = 2L,$$

it follows that

$$\bar{k} = \frac{2L}{N}.$$

This formula is fundamental. It says that the average degree is determined entirely by the number of nodes and the number of links.

A useful interpretation is the following:

- each link contributes 1 to the degree of one endpoint
- and also contributes 1 to the degree of the other endpoint
- so each link contributes a total of 2 to the sum of all node degrees

That is why the numerator is $2L$ rather than L.

2.6.1 A quick numerical example

Suppose a graph has

$$N = 10 \qquad \text{and} \qquad L = 15.$$

Then the average degree is

$$\bar{k} = \frac{2L}{N} = \frac{2 \times 15}{10} = 3.$$

This does **not** mean that every node has degree 3. It only means that the average over all node degrees is 3.

For example, one graph could have degrees

$$(1, 2, 2, 3, 3, 3, 4, 4, 4, 4),$$

whose average is 3. Another graph with a very different structure might still have the same average degree. So average degree is useful, but it does not capture the full shape of the degree distribution.

2.6.2 Why average degree matters in random graphs

In a fixed graph, $\bar{k}$ is a property of that one graph. In a random graph model, the situation is different because the graph itself is random. That means the number of links may vary from one realization to another, and therefore the average degree may also vary.

So in a random graph ensemble, the natural quantity to study is the **expected average degree**,

$$\mathbb{E}[\bar{k}].$$

This is the average value of $\bar{k}$ over all graphs generated by the model.

Starting from

$$\bar{k} = \frac{2L}{N},$$

and taking expectation on both sides gives

$$\mathbb{E}[\bar{k}] = \mathbb{E}\left[\frac{2L}{N}\right].$$

Since N is fixed in both Erdős–Rényi models, the constant $\frac{2}{N}$ can be taken outside the expectation:

$$\mathbb{E}[\bar{k}] = \frac{2\,\mathbb{E}[L]}{N}.$$

This relation is extremely useful. It says that once the expected number of links is known, the expected average degree follows immediately.

> **Key relation**
>
> For any random graph ensemble with fixed number of nodes N,
>
> $$\mathbb{E}[\bar{k}] = \frac{2\,\mathbb{E}[L]}{N}.$$
>
> So the problem of expected average degree reduces to the problem of expected number of links.

2.6.3 Average degree in $G(N, L)$

In the model $G(N, L)$, the number of links is fixed from the beginning. Every graph in the ensemble has exactly L links.

That means there is no randomness in the value of L itself:

$$\mathbb{E}[L] = L.$$

Substituting this into the general relation gives

$$\mathbb{E}[\bar{k}] = \frac{2L}{N}.$$

So in $G(N, L)$, the average degree is not only expected to equal $\frac{2L}{N}$; in fact every graph in the ensemble has that same average degree.

This is an important distinction. The detailed arrangement of edges changes from one graph to another, but the total number of edges does not. Therefore, the average degree is fixed across the whole ensemble.

2.6.3.1 Interpretation in $G(N, L)$

The model $G(N, L)$ allows randomness in **structure**, but not in **density**.

- which nodes are connected is random
- how many total links exist is fixed
- therefore the average degree is fixed

For example, in $G(6, 4)$ the average degree is always

$$\bar{k} = \frac{2 \times 4}{6} = \frac{8}{6} = \frac{4}{3}.$$

Different realizations may have very different shapes, but all of them must share this same average degree.

2.6.4 Average degree in $G(N, p)$

The model $G(N, p)$ is different because the total number of links is no longer fixed in advance.

There are

$$K = \binom{N}{2}$$

possible undirected edges, and each one is included independently with probability p.

So the total number of links L is a binomial random variable:

$$L \sim B(K, p).$$

This immediately gives

$$\mathbb{E}[L] = pK.$$

Substituting into the general relation,

$$\mathbb{E}[\bar{k}] = \frac{2\,\mathbb{E}[L]}{N} = \frac{2pK}{N}.$$

Now replace K by $\binom{N}{2}$:

$$\mathbb{E}[\bar{k}] = \frac{2p}{N}\binom{N}{2}.$$

Since

$$\binom{N}{2} = \frac{N(N-1)}{2},$$

this simplifies to

$$\mathbb{E}[\bar{k}] = \frac{2p}{N} \cdot \frac{N(N-1)}{2} = p(N-1).$$

So the expected average degree in $G(N, p)$ is

$$\mathbb{E}[\bar{k}] = p(N-1).$$

2.6.5 A second way to understand the formula

The formula

$$\mathbb{E}[\bar{k}] = p(N-1)$$

can also be understood directly, without first going through $\mathbb{E}[L]$.

Take one node. It has $N-1$ possible neighbors. For each of those possible neighbors, an edge is present with probability p.

So the expected number of neighbors of that node is simply

$$(N-1) \times p = p(N-1).$$

Because all nodes are statistically equivalent in the Erdős–Rényi model, this is also the expected average degree of the whole graph.

This second interpretation is very useful because it makes the formula intuitive:

- $N-1$ = number of possible neighbors
- p = chance that each possible link exists
- expected degree = "number of opportunities" times "probability of success"

2.6.6 A quick numerical example in $G(N, p)$

Suppose

$$N = 50 \qquad \text{and} \qquad p = 0.10.$$

Then

$$\mathbb{E}[\bar{k}] = p(N-1) = 0.10 \times 49 = 4.9.$$

So a typical graph in this ensemble should have an average degree close to 4.9, although the exact value will vary from one realization to another.

This illustrates an important contrast with $G(N, L)$:

- in $G(N, L)$ the average degree is exactly fixed
- in $G(N, p)$ the average degree fluctuates, but its expected value is fixed

2.6.7 Dense and sparse regimes

The formula

$$\mathbb{E}[\bar{k}] = p(N-1)$$

also helps explain how the graph changes when N or p changes.

If p is fixed and N increases, then the expected average degree grows roughly linearly with N. In that case the graph becomes denser as the system grows.

On the other hand, many real network settings are better modeled by keeping the expected degree moderate even when N becomes large. To do that, p must decrease as N grows. This is exactly the sparse regime that later leads to the Poisson approximation for the degree distribution.

So the average degree is not just a descriptive quantity. It also helps identify whether the graph is sparse or dense.

2.6.8 Comparing the two models

The formulas for the two Erdős–Rényi models are:

$$\mathbb{E}[\bar{k}] = \frac{2L}{N} \qquad \text{for } G(N, L),$$

and

$$\mathbb{E}[\bar{k}] = p(N-1) \qquad \text{for } G(N, p).$$

These two expressions describe the same idea in two different parameterizations:

- in $G(N, L)$ the model is controlled by the number of links L
- in $G(N, p)$ the model is controlled by the edge probability p

They are closely related. In fact, if a graph from $G(N, p)$ has expected number of links

$$\mathbb{E}[L] = p\binom{N}{2},$$

then it is natural to compare it with a graph from $G(N, L)$ using approximately that same value of L.

2.6.9 Summary

The average degree is one of the simplest and most useful global quantities in random graph theory.

The main points are:

- in any graph with N nodes and L links,
$$\bar{k} = \frac{2L}{N}$$
- in a random graph ensemble,
$$\mathbb{E}[\bar{k}] = \frac{2\,\mathbb{E}[L]}{N}$$
- in $G(N, L)$,
$$\mathbb{E}[\bar{k}] = \frac{2L}{N}$$
- in $G(N, p)$,
$$\mathbb{E}[\bar{k}] = p(N-1)$$

So the expected average degree gives a first measure of how densely connected a random graph is, and it provides a natural bridge between the combinatorial description of the model and its probabilistic behavior.

2.7 Degree distribution

The degree distribution describes how node degrees are distributed across a graph. In this chapter, the important point is not the definition of degree itself, but how a random graph model gives rise to a distribution of degrees across nodes.

A useful way to introduce this idea is to move in three steps:

1. start from one concrete graph
2. record the degree of each node
3. transform those degree values into a distribution

2.7.1 A small example

Consider the following graph.

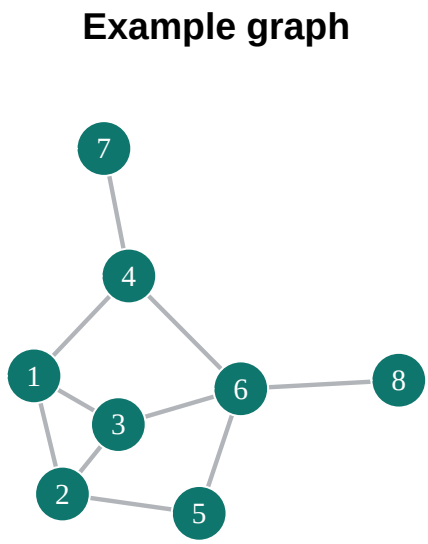


Figure 2.3: A small graph used to build the degree distribution step by step.

For this graph, the degree of each node is:

Node	Degree
1	3
2	3
3	3
4	3
5	2
6	4
7	1
8	1

This table contains node-level information. It tells the degree of each specific node, but it does not yet summarize how the degree values are distributed across the graph.

2.7.2 From node degrees to degree counts

The next step is to count how many nodes share each degree value.

Degree k	Number of nodes N_k	Proportion $p_k = \frac{N_k}{N}$
0	0	0
1	2	0.25
2	1	0.125

Degree k	Number of nodes N_k	Proportion $p_k = \frac{N_k}{N}$
3	4	0.50
4	1	0.125

This table is the key transition from node-level data to a graph-level summary.

- the first column gives the degree value k
- the second column gives the number of nodes having that degree
- the third column gives the fraction of nodes having that degree

If the graph has N nodes in total and N_k nodes of degree k, then the empirical degree distribution is

$$p_k = \frac{N_k}{N}.$$

So p_k is the proportion of nodes whose degree is exactly k.

2.7.3 Plotting the degree distribution

The same information can be shown visually.

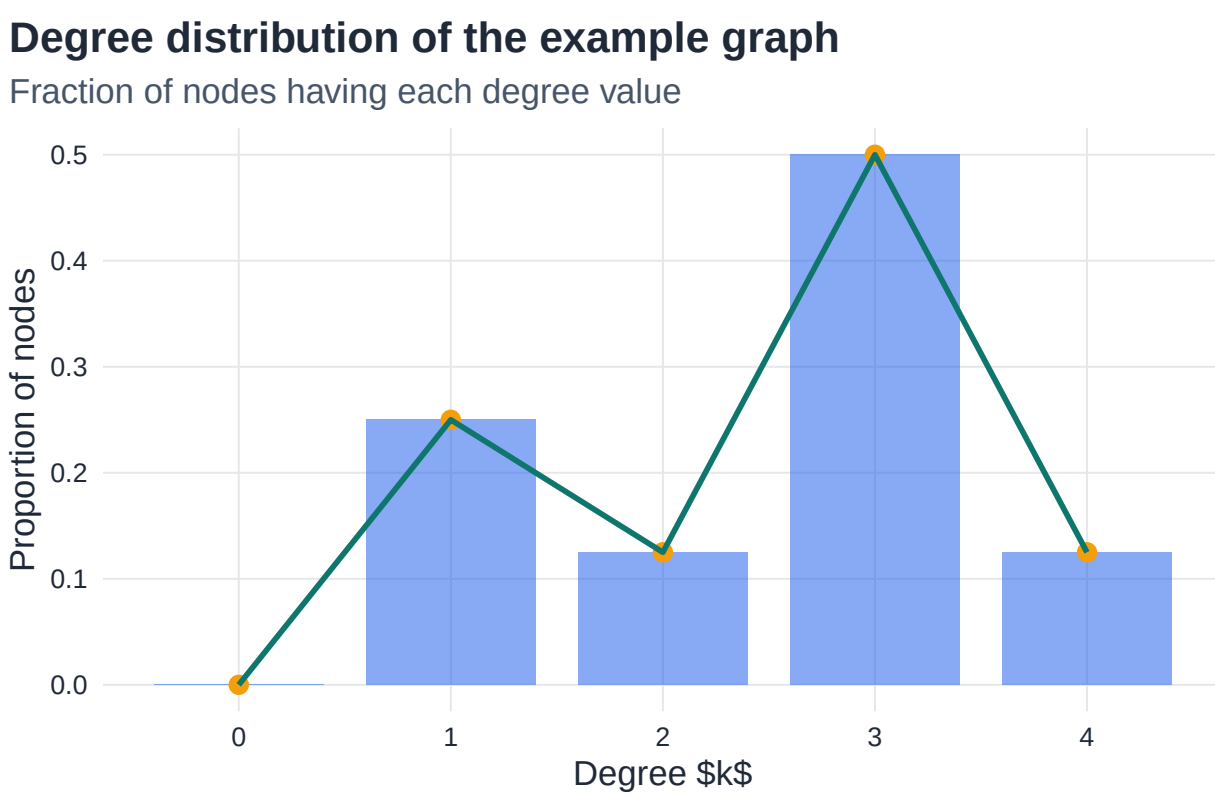


Figure 2.4: Degree distribution of the example graph.

This plot shows the degree distribution in its most direct form:

- the horizontal axis gives the degree value k
- the vertical axis gives the proportion of nodes with that degree

The distinction between two ideas is now clear:

- the **degree table** records the degree of each node
- the **degree distribution** records how common each degree value is in the graph

2.7.4 Degree distribution in the random graph model $G(N, p)$

Now move from one fixed graph to the random graph model $G(N, p)$.

Take one node. It has $N-1$ possible neighbors. For each possible neighbor, an edge is present with probability p and absent with probability $1-p$. Because these edge decisions are independent, the degree of that node is the number of successes in $N-1$ Bernoulli trials. Therefore,

$$k \sim B(N-1, p).$$

Hence the probability that a node has degree exactly k is

$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}.$$

This is the degree distribution of the Erdős–Rényi model $G(N, p)$.

2.7.5 Interpreting the formula

Each factor in the formula has a clear meaning.

To obtain degree k for a chosen node:

1. exactly k of the $N-1$ possible edges must be present
2. the remaining $N-1-k$ edges must be absent
3. there are $\binom{N-1}{k}$ different ways to choose which k neighbors are connected

Therefore,

$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}.$$

The three parts of this expression play different roles:

- $\binom{N-1}{k}$ counts the possible choices of the k neighbors
- p^k gives the probability that those k edges are present
- $(1-p)^{N-1-k}$ gives the probability that the remaining edges are absent

The expected value of this distribution is

$$\mathbb{E}[k] = p(N-1),$$

which agrees with the expected average degree derived earlier.

2.7.6 Comparing one realization with the theoretical distribution

The formula above describes the degree distribution predicted by the model. A single simulated graph from $G(N, p)$ will fluctuate around that prediction.

```r
n <- 80
p <- 0.08

g <- sample_gnp(n = n, p = p, directed = FALSE, loops = FALSE)
obs_deg <- degree(g)

obs_tbl <- tibble(k = obs_deg) |>
  count(k, name = "observed") |>
  complete(k = 0:(n - 1), fill = list(observed = 0)) |>
  mutate(observed_prob = observed / sum(observed))

theory_tbl <- tibble(k = 0:(n - 1)) |>
  mutate(theory_prob = dbinom(k, size = n - 1, prob = p))

plot_tbl <- left_join(obs_tbl, theory_tbl, by = "k")

ggplot(plot_tbl, aes(x = k)) +
  geom_col(aes(y = observed_prob), fill = course_cols$blue, alpha = 0.45, width = 0.82) +
  geom_line(aes(y = theory_prob), color = course_cols$teal, linewidth = 1.1) +
  geom_point(aes(y = theory_prob), color = course_cols$gold, size = 2.2) +
  labs(
    title = "Observed and theoretical degree distribution",
    subtitle = sprintf("Single realization of G(%d, %.2f)", n, p),
    x = "Degree $k$",
    y = "Probability"
  )
```

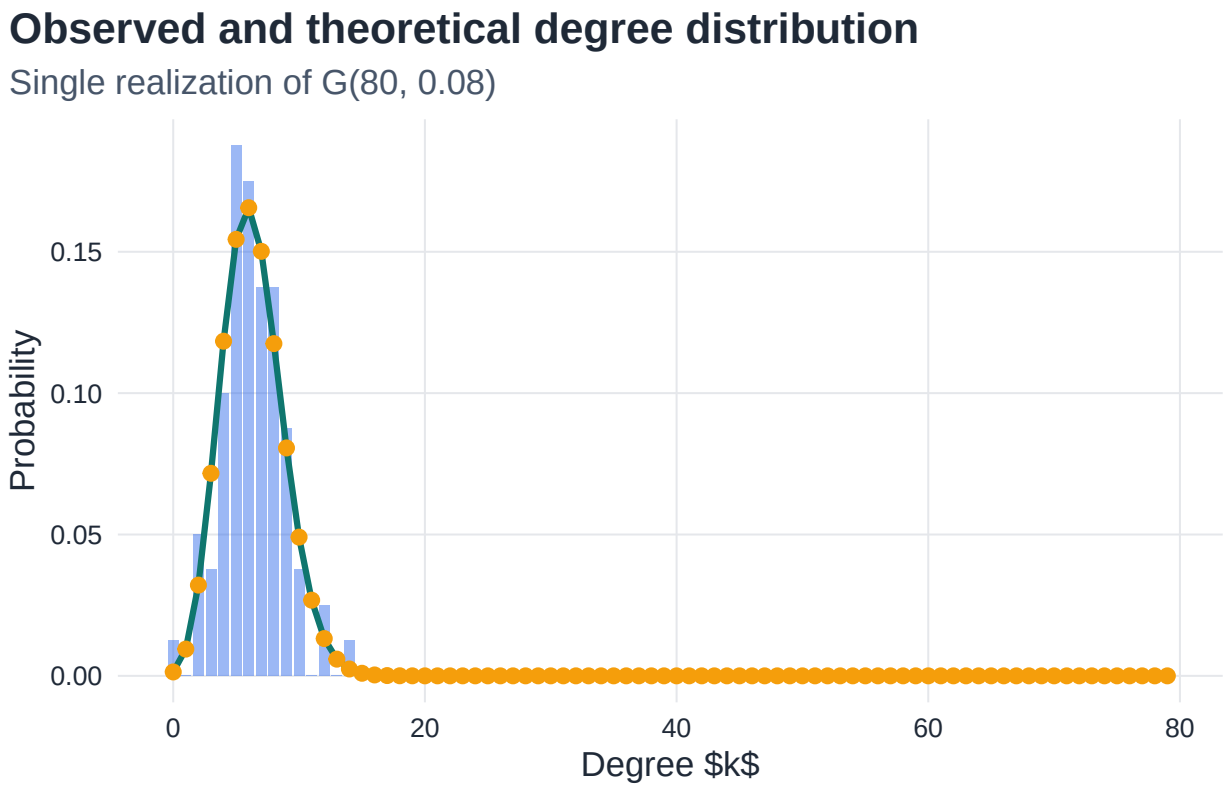


Figure 2.5: Observed degree frequencies in one simulated graph compared with the theoretical degree distribution in G(N,p).

The bars show the empirical degree distribution of one generated graph. The line and points show the theoretical distribution

$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}.$$

The two are not expected to coincide perfectly in one realization. The figure shows how the observed degree frequencies of a random graph relate to the probability law defined by the model.

! Summary

In this chapter, the degree distribution can be read at three connected levels:

1. a graph has nodes with different degree values
2. those degree values can be summarized by the empirical distribution

$$p_k = \frac{N_k}{N}$$

3. in the model $G(N, p)$, the degree of a node follows the binomial law

$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}$$

2.8 Poisson approximation in sparse large graphs

The exact binomial formula is correct, but in many network settings the focus is on large graphs that are relatively sparse.

A graph is called **sparse** when the number of links grows much more slowly than the maximum possible number of links. In the Erdős–Rényi setting, a common sparse regime is:

- N is large
- p is small
- the mean degree remains finite

Let

$$\lambda = p(N-1).$$

If N is large and p is small in such a way that λ stays finite, then the binomial degree distribution is well approximated by a Poisson distribution:

$$P(k) \approx e^{-\lambda} \frac{\lambda^k}{k!}.$$

2.8.1 Why this approximation appears

This approximation reflects a common probabilistic pattern:

- each individual edge is unlikely
- there are many possible edges
- the total number of successes remains moderate

That is exactly the setting in which Poisson limits naturally appear.

This is an important modeling lesson. The Poisson distribution is not introduced here as a completely different idea. It appears as a limit of the binomial degree distribution in the sparse large-N regime.

> Interpretation
>
> In a large sparse Erdős–Rényi graph, most nodes have relatively small degree, and the degree distribution becomes sharply concentrated around a finite mean.

2.8.2 Class example: binomial distribution and Poisson approximation

```
n <- 300
p <- 0.02
lambda <- p * (n - 1)
k_vals <- 0:24

compare_tbl <- tibble(
  k = k_vals,
  binomial = dbinom(k_vals, size = n - 1, prob = p),
  poisson = dpois(k_vals, lambda = lambda)
) |>
  pivot_longer(cols = c(binomial, poisson), names_to = "distribution", values_to = "probability") |>
  mutate(distribution = recode(distribution, binomial = "Binomial", poisson = "Poisson"))

ggplot(compare_tbl, aes(x = k, y = probability, color = distribution)) +
```

```r
  geom_line(linewidth = 1.1) +
  geom_point(size = 2.2) +
  scale_color_manual(values = c("Binomial" = course_cols$teal, "Poisson" = course_cols$coral)) +
  labs(
    title = "Binomial and Poisson degree distributions",
    subtitle = sprintf("n = %d, p = %.3f, mean degree = %.2f", n, p, lambda),
    x = "Degree $k$",
    y = "Probability",
    color = NULL
  )
```

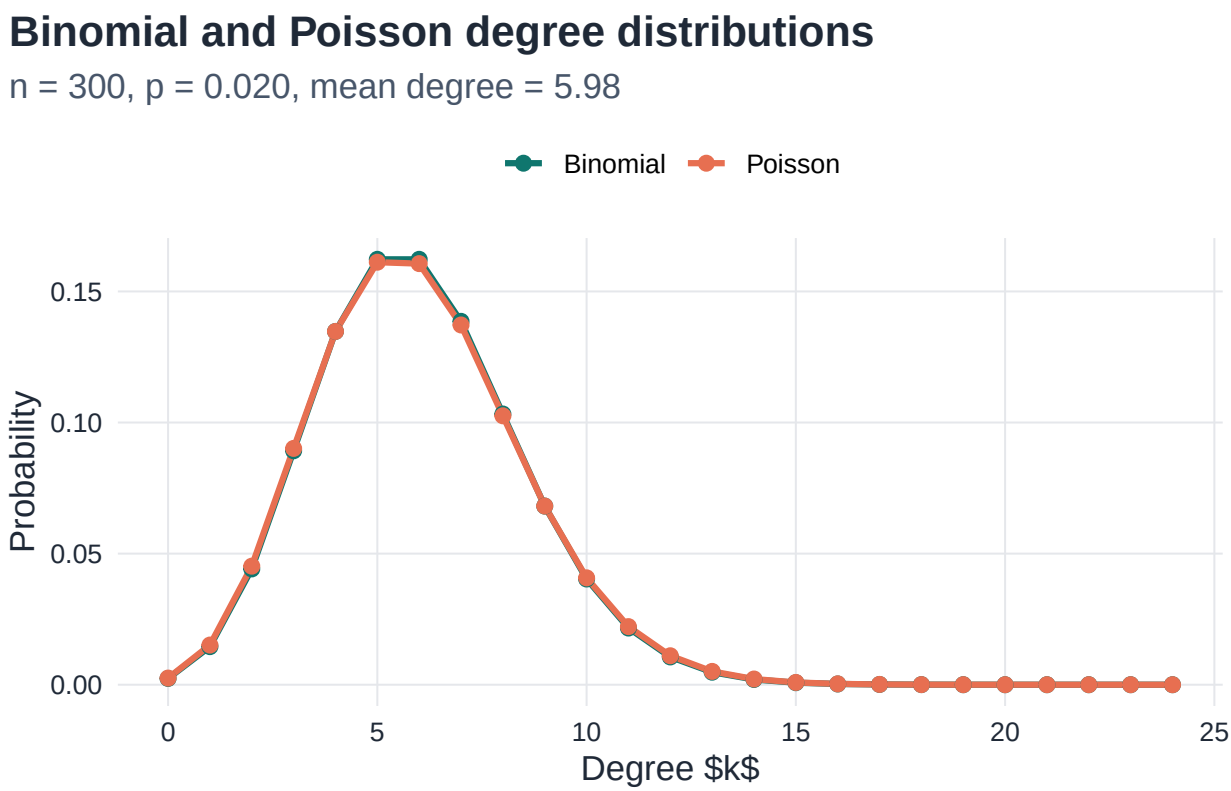


Figure 2.6: Comparison between the exact binomial degree distribution and its Poisson approximation.

2.9 Clustering coefficient

The clustering coefficient measures how strongly the neighbors of a node are connected to one another. In other words, once a node is chosen, clustering asks whether its neighbors also tend to be neighbors of each other.

This idea is important because two graphs can have similar size, similar average degree, and even similar degree distributions, while still differing strongly in how much local triangle structure they contain. The clustering coefficient is one of the main quantities used to capture that local structure.

2.9.1 Local clustering coefficient

Suppose node i has degree k_i. Then node i has exactly k_i neighbors.

Among those k_i neighbors, the maximum possible number of edges is

$$L_i^{\max} = \binom{k_i}{2} = \frac{k_i(k_i-1)}{2}.$$

If the actual number of edges among those neighbors is L_i, then the local clustering coefficient of node i is

$$C_i = \frac{L_i}{L_i^{\max}} = \frac{2L_i}{k_i(k_i-1)}.$$

This formula compares:

- the number of neighbor-neighbor edges that actually exist
- the maximum number of neighbor-neighbor edges that could exist

So C_i is always between 0 and 1:

- $C_i = 0$ means none of the neighbors of i are connected to each other
- $C_i = 1$ means all neighbors of i are mutually connected
- intermediate values indicate partial local connectivity

2.9.2 A fixed node with three different neighborhood structures

To see how this works clearly, keep the same central node A and the same three neighbors B, C, and D, and only change the edges among the neighbors.

In all three examples below:

- the focal node is A
- the neighbors of A are B, C, and D
- therefore $k_A = 3$
- the maximum possible number of edges among the neighbors is

$$L_A^{\max} = \binom{3}{2} = 3$$

So the only question is: how many of the three possible neighbor-neighbor edges actually exist?

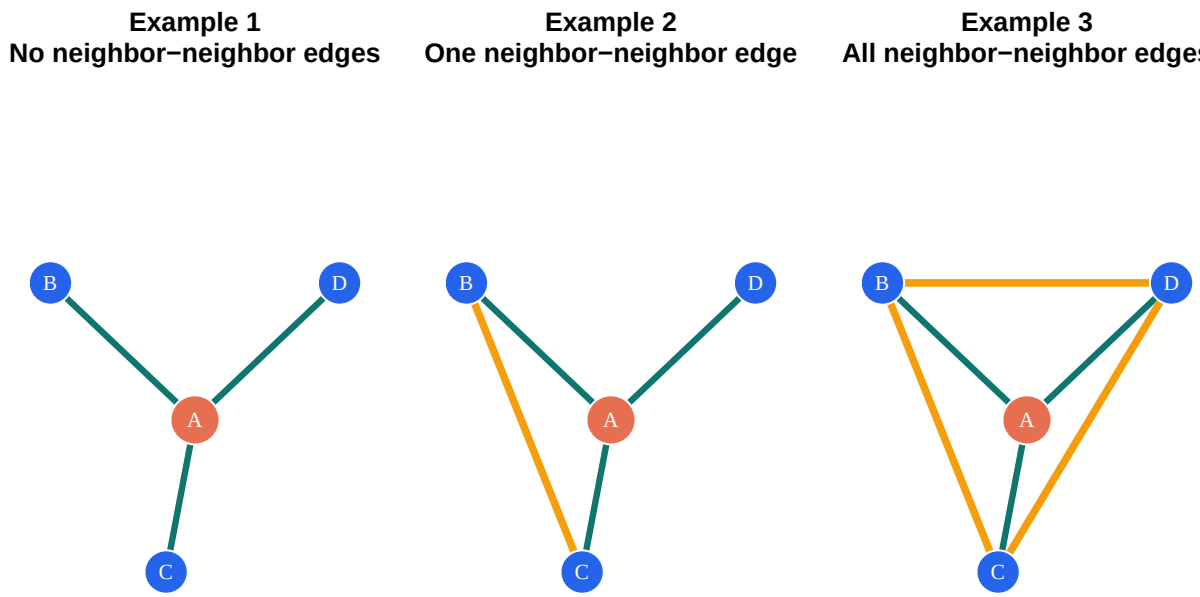


Figure 2.7: Three example neighborhoods for the same focal node A. Teal edges connect A to its neighbors; gold edges connect neighbors to each other and contribute to clustering.

In these figures:

- the **teal edges** connect the focal node A to its neighbors
- the **gold edges** connect neighbors to each other
- only the gold edges count in the value of L_A

2.9.3 Example 1

In the first graph, node A has neighbors B, C, and D, but none of those neighbors are connected to each other.

So

$$L_A = 0 \qquad \text{and} \qquad L_A^{\max} = 3.$$

Therefore,

$$C_A = \frac{L_A}{L_A^{\max}} = \frac{0}{3} = 0.$$

This is the smallest possible clustering value. The neighbors of A form no local triangle structure at all.

2.9.4 Example 2

In the second graph, the same three neighbors are present, but now one edge exists among them, namely between B and C.

So

$$L_A = 1 \qquad \text{and} \qquad L_A^{\max} = 3.$$

Therefore,

$$C_A = \frac{L_A}{L_A^{\max}} = \frac{1}{3}.$$

This means that only one-third of the possible neighbor-neighbor edges are present.

2.9.5 Example 3

In the third graph, all three neighbor-neighbor edges are present:

$$(B, C), \qquad (B, D), \qquad (C, D).$$

So

$$L_A = 3 \qquad \text{and} \qquad L_A^{\max} = 3.$$

Therefore,

$$C_A = \frac{L_A}{L_A^{\max}} = \frac{3}{3} = 1.$$

This is the largest possible clustering value. The neighbors of A form a complete subgraph, so the local neighborhood is maximally clustered.

2.9.6 Summary table for the three examples

Example	Degree of focal node k_A	Maximum possible neighbor-neighbor edges $L_A^{\max}$	Actual neighbor-neighbor edges L_A	Clustering coefficient C_A
1	3	3	0	0
2	3	3	1	$\frac{1}{3}$
3	3	3	3	1

This table makes the logic of the formula very explicit. The degree stays the same in all three examples. What changes is only the number of links among the neighbors.

2.9.7 What clustering is really measuring

These examples show an important point:

- degree counts how many neighbors a node has
- clustering measures how tightly those neighbors are connected to each other

So two nodes can have the same degree but very different clustering coefficients.

For example, in all three graphs above, node A has degree 3. But its clustering coefficient changes from

$$0 \quad \text{to} \quad \frac{1}{3} \quad \text{to} \quad 1.$$

So clustering reveals local structural information that degree alone cannot capture.

2.9.8 Average clustering coefficient of a graph

Once the local clustering coefficient has been defined for every node, the average clustering coefficient of the graph is

$$\bar{C} = \frac{1}{N} \sum_{i=1}^{N} C_i.$$

This quantity measures the overall tendency of the graph to form locally connected neighborhoods.

It should be interpreted carefully:

- C_i is a node-level quantity
- $\bar{C}$ is a graph-level average

The local quantity tells how clustered one neighborhood is. The average quantity summarizes the graph as a whole.

2.9.9 Why clustering is important in network science

Clustering is especially important because many real networks are rich in triangle structure.

In social networks, for example, if one person knows two others, there is often an elevated chance that those two people also know each other. This kind of local closure is one of the characteristic patterns of social graphs.

Random graphs of the Erdős–Rényi type behave differently. Since edges are placed independently, there is no special mechanism that encourages triangle closure beyond the baseline edge probability itself.

That is why clustering becomes an important comparison point between random graphs and real-world networks.

2.9.10 Expected clustering in $G(N, p)$

Now consider the model $G(N, p)$.

Suppose node i has some set of neighbors. To compute its clustering coefficient, the question is whether pairs among those neighbors are connected.

In the Erdős–Rényi model, every possible edge appears independently with probability p. Therefore, once two neighbors of node i are chosen, the probability that they are connected is simply p.

So the expected fraction of connected neighbor-neighbor pairs is

$$\mathbb{E}[C_i] = p.$$

As a consequence, the expected average clustering coefficient is also

$$\mathbb{E}[\bar{C}] = p.$$

This is a very simple and very important result.

> **Key result for Erdős–Rényi graphs**
>
> In the model $G(N, p)$, the expected clustering coefficient is
>
> $$\mathbb{E}[C] = p.$$
>
> So clustering is directly controlled by the edge probability.

2.9.11 Why this result matters

This formula shows that Erdős–Rényi graphs usually have low clustering when p is small.

That is one of the main reasons why Erdős–Rényi graphs are useful as a baseline but often unrealistic as full models of social networks:

- they capture random edge formation well
- but they do not create unusually strong local triangle closure

So if a real network has clustering much larger than p, that indicates structure beyond independent random linking.

2.9.12 Numerical check by simulation

The theoretical prediction in $G(N, p)$ is

$$\mathbb{E}[\bar{C}] = p.$$

This can be checked numerically by simulating many graphs and computing the average clustering coefficient in each one.

```r
n <- 100
p <- 0.07
reps <- 250

sim_tbl <- map_dfr(seq_len(reps), function(i) {
  g <- sample_gnp(n = n, p = p, directed = FALSE, loops = FALSE)
  tibble(
    sim = i,
    avg_degree = mean(degree(g)),
    avg_clustering = transitivity(g, type = "average")
  )
})

summary_tbl <- tibble(
  quantity = c("Average degree", "Average clustering"),
  empirical_mean = c(mean(sim_tbl$avg_degree), mean(sim_tbl$avg_clustering)),
```

```r
  theoretical_value = c(p * (n - 1), p)
)

summary_tbl
```

```
# A tibble: 2 x 3
  quantity           empirical_mean theoretical_value
  <chr>                       <dbl>             <dbl>
1 Average degree             6.93               6.93
2 Average clustering         0.0706             0.07
```

```r
ggplot(sim_tbl, aes(x = avg_clustering)) +
  geom_histogram(bins = 24, fill = course_cols$teal, alpha = 0.65, color = "white") +
  geom_vline(xintercept = p, color = course_cols$gold, linewidth = 1.1, linetype = 2) +
  labs(
    title = "Average clustering across simulations",
    subtitle = sprintf("Dashed line = theoretical value p = %.2f", p),
    x = "Average clustering",
    y = "Count"
  )
```

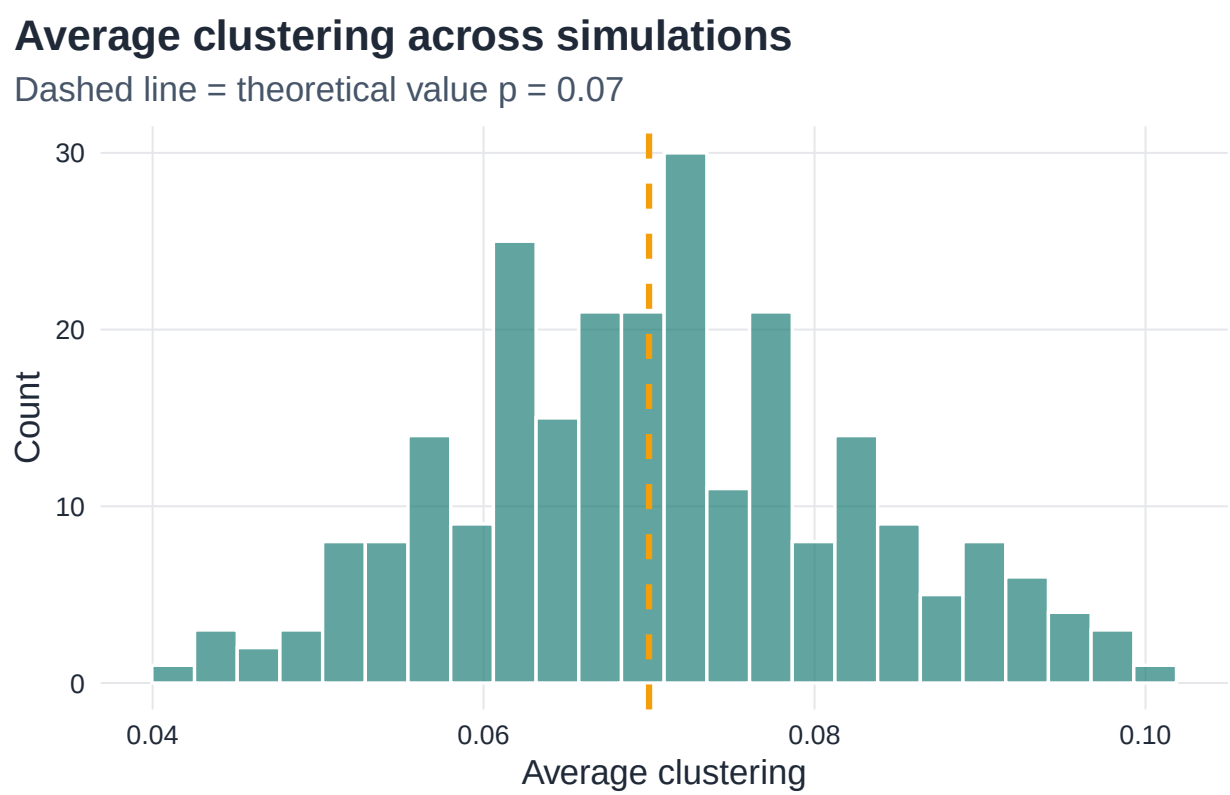


Figure 2.8: Average clustering across repeated simulations with the theoretical value shown as a dashed line.

The histogram shows how the observed average clustering varies across realizations of the model, while the dashed line marks the theoretical value p.

2.9.13 Summary

The clustering coefficient adds information that degree alone cannot provide.

The main points are:

- for a node i with degree k_i, the maximum number of possible edges among its neighbors is

$$\binom{k_i}{2}$$

- the local clustering coefficient is

$$C_i = \frac{2L_i}{k_i(k_i - 1)}$$

- it measures how strongly the neighbors of a node are connected to one another
- the graph-level average is

$$\bar{C} = \frac{1}{N}\sum_{i=1}^{N} C_i$$

- in the Erdős–Rényi model $G(N, p)$,

$$\mathbb{E}[C] = p$$

So clustering is a local structural measure that becomes especially useful when comparing random graphs with more realistic network models. ## Core formulas to remember

Summary

For Erdős–Rényi random graphs:

1. Number of possible undirected edges:

$$K = \binom{N}{2}$$

2. Probability of a graph in $G(N, L)$:

$$P(G) = \frac{1}{\binom{K}{L}}$$

3. Probability of a graph with L links in $G(N, p)$:

$$P(G) = p^L(1-p)^{K-L}$$

4. Expected average degree in $G(N, p)$:

$$\mathbb{E}[\bar{k}] = p(N-1)$$

5. Degree distribution in $G(N, p)$:

$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}$$

6. Poisson approximation in the sparse regime:

$$P(k) \approx e^{-\lambda}\frac{\lambda^k}{k!}, \qquad \lambda = p(N-1)$$

7. Expected clustering in $G(N, p)$:

$$\mathbb{E}[C] = p$$

2.10 Class example: generate and plot a random graph

```
n <- 20
p <- 0.12

g_er <- sample_gnp(n = n, p = p, directed = FALSE, loops = FALSE)

plot(
  g_er,
  layout = layout_with_fr(g_er),
  vertex.size = 22,
  vertex.color = course_cols$teal,
  vertex.frame.color = "white",
  vertex.label.color = "white",
  edge.color = scales::alpha(course_cols$ink, 0.32),
  vertex.label.cex = 0.8,
  main = sprintf("A realization of G(%d, %.2f)", n, p)
)
```

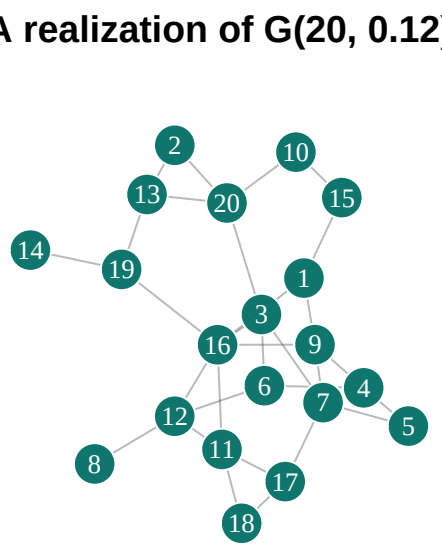
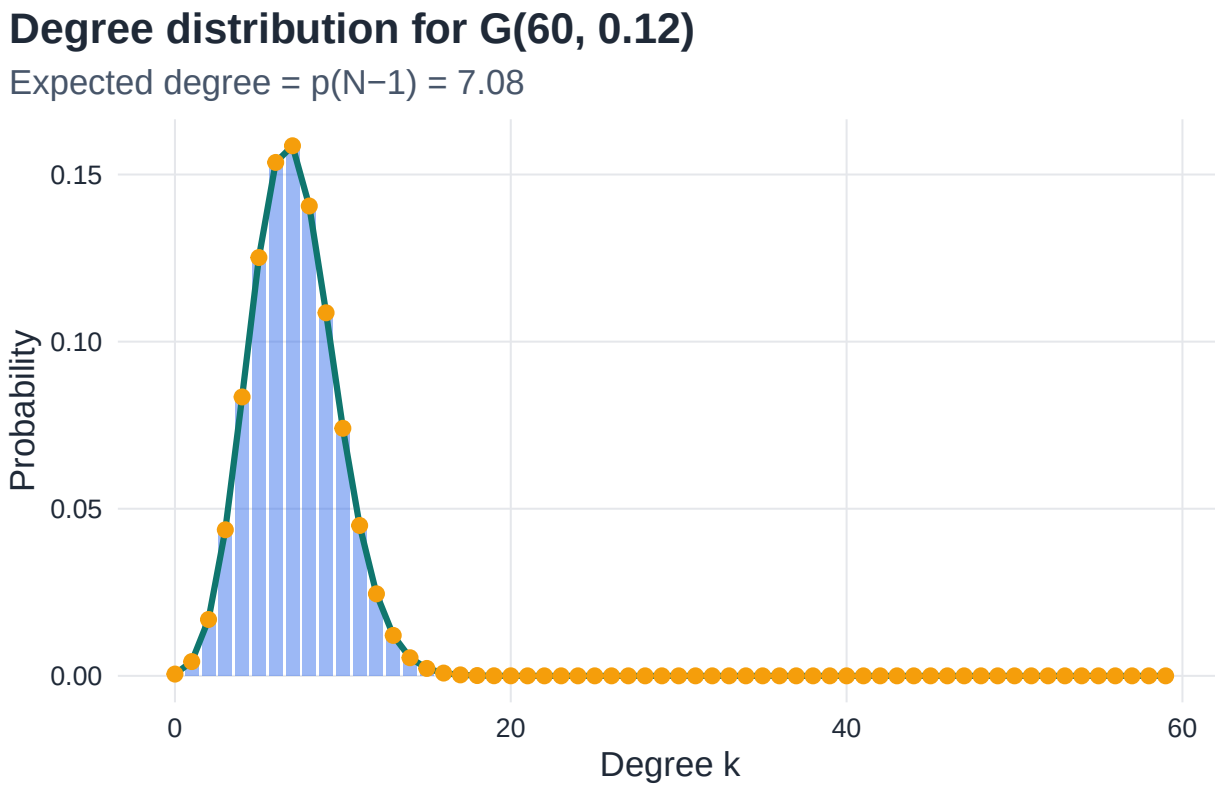


Figure 2.9: One realization of an Erdős–Rényi graph.

2.11 Interactive explorer

In the PDF version, the interactive widget is replaced by a static plot.



2.12 In-class questions

1. Why is a random graph a model rather than a single graph?
2. What is the main conceptual difference between $G(N, L)$ and $G(N, p)$?
3. Why does the total number of edges follow a binomial distribution in $G(N, p)$?
4. Why does the degree of a single node also follow a binomial distribution?
5. What does sparse mean in this chapter?
6. Why does the Poisson approximation appear in sparse large graphs?
7. Why is clustering in Erdős–Rényi graphs usually lower than in social networks?

2.13 End-of-chapter exercises

The exercises below are divided into three groups:

1. mathematical exercises,
2. conceptual and analytical exercises,
3. simulation and coding exercises.

They are designed to move from formal derivation to interpretation and numerical verification.

2.13.1 Part I. Mathematical exercises

2.13.1.1 Exercise 1. Counting graphs in $G(N, L)$

Let $G(N, L)$ be the Erdős–Rényi ensemble with N vertices and exactly L edges.

1. Show that the number of possible undirected edges is

$$K = \binom{N}{2}.$$

2. Prove that the number of labeled simple graphs in the ensemble is

$$\Omega(N, L) = \binom{K}{L}.$$

3. Deduce that the probability of any one graph in the ensemble is

$$P(G) = \frac{1}{\binom{K}{L}}.$$

2.13.1.2 Exercise 2. Counting graphs in $G(N, p)$

Let $G(N, p)$ be the Erdős–Rényi ensemble in which every possible edge is present independently with probability p.

1. Let a graph G contain exactly L edges. Show that its probability is

$$P(G) = p^L (1-p)^{K-L}, \qquad K = \binom{N}{2}.$$

2. Prove that all graphs with the same number of edges have the same probability.
3. Explain why graphs with different numbers of edges generally do not have the same probability.

2.13.1.3 Exercise 3. Edge probability in $G(N, L)$

In the ensemble $G(N, L)$, compute the probability that two fixed distinct vertices are connected by an edge.

Your derivation should start from combinatorial counting and end with a simplified formula in terms of N and L.

2.13.1.4 Exercise 4. Average degree in $G(N, L)$

Let $\bar{k}$ denote the average degree of a graph.

1. Prove that for every graph with N vertices and L edges,

$$\bar{k} = \frac{2L}{N}.$$

2. Deduce that in the ensemble $G(N, L)$,

$$\mathbb{E}[\bar{k}] = \frac{2L}{N}.$$

3. Explain why this quantity is not random in the ensemble $G(N, L)$.

2.13.1.5 Exercise 5. Average degree in $G(N,p)$

In the ensemble $G(N,p)$:

1. Show that the total number of edges satisfies
$$L \sim B\left(\binom{N}{2}, p\right).$$
2. Deduce that
$$\mathbb{E}[L] = p\binom{N}{2}.$$
3. Prove that
$$\mathbb{E}[\bar{k}] = p(N-1).$$

2.13.1.6 Exercise 6. Degree distribution in $G(N,p)$

Fix a vertex v in the model $G(N,p)$.

1. Show that the degree of v is a binomial random variable.
2. Derive the formula
$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}.$$
3. Compute its mean and variance.

2.13.1.7 Exercise 7. Poisson approximation

Assume that N is large, p is small, and

$$\lambda = p(N-1)$$

remains finite.

1. Starting from the exact degree distribution in $G(N,p)$,
$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k},$$
derive the Poisson approximation
$$P(k) \approx e^{-\lambda} \frac{\lambda^k}{k!}.$$
2. State clearly which approximations you use.
3. Explain why this approximation is natural in the sparse regime.

2.13.1.8 Exercise 8. Clustering coefficient in $G(N,p)$

Let i be a vertex of degree k_i.

1. Show that the maximum possible number of edges among the neighbors of i is
$$\binom{k_i}{2}.$$

2. Starting from the definition of local clustering coefficient, show that in $G(N, p)$,

$$\mathbb{E}[C_i] = p.$$

3. Deduce that

$$\mathbb{E}[\bar{C}] = p.$$

2.13.1.9 Exercise 9. Directed version of the models

Generalize the definitions of $G(N, L)$ and $G(N, p)$ to directed graphs without self-loops.

1. Determine the total number of possible directed edges.
2. Write the probability of a directed graph in the $G(N, L)$ version.
3. Write the probability of a directed graph in the $G(N, p)$ version.
4. Derive the expected in-degree and expected out-degree.

2.13.1.10 Exercise 10. Most probable graph size in $G(N, p)$

For fixed N and fixed p, the probability that a graph has exactly L edges is

$$P(L) = \binom{K}{L} p^L (1-p)^{K-L}, \qquad K = \binom{N}{2}.$$

Determine the value(s) of L that maximize this probability.

2.13.2 Part II. Conceptual and analytical exercises

2.13.2.1 Exercise 11. $G(N, L)$ versus $G(N, p)$

Write a short analytical comparison between the two models. Your answer should address:

- what is fixed in each model,
- what is random in each model,
- what it means for graphs to be equiprobable,
- why the two models are related but not identical.

2.13.2.2 Exercise 12. Same average degree, different structure

Construct two labeled graphs on the same number of vertices such that they have the same average degree but visibly different structure.

Discuss why average degree alone is insufficient to characterize a graph.

2.13.2.3 Exercise 13. Degree distribution versus average degree

Explain why knowing only $\mathbb{E}[\bar{k}]$ does not determine the degree distribution.

Your answer should include: - one statement about what average degree measures, - one statement about what the degree distribution measures, - one example showing that different degree distributions may have the same average degree.

2.13.2.4 Exercise 14. Why Erdős–Rényi clustering is small

Give a mathematical and conceptual explanation of why clustering in an Erdős–Rényi graph is usually low when p is small.

Your answer should distinguish between: - independent edge formation, - local triangle closure, - the behavior of social networks.

2.13.2.5 Exercise 15. Sparse versus dense regimes

Discuss the difference between the sparse and dense regimes in $G(N, p)$.

Your answer should address: - how p scales with N, - how $\mathbb{E}[k]$ behaves, - why the Poisson approximation belongs to the sparse regime.

2.13.2.6 Exercise 16. Why the Poisson law is only an approximation

The degree distribution in $G(N, p)$ is exactly binomial, but often approximated by a Poisson law.

Explain why the Poisson distribution is not exact in general, and state clearly the regime in which it becomes accurate.

2.13.3 Part III. Simulation and coding exercises

2.13.3.1 Exercise 17. Simulating $G(N, p)$ and estimating the degree distribution

Write an `R` function that:

1. generates m independent graphs from $G(N, p)$,
2. computes the empirical degree distribution across all simulated graphs,
3. compares it with the theoretical binomial distribution.

Use the function for several values of N and p.

2.13.3.2 Exercise 18. Binomial versus Poisson comparison

For a sequence of values of N and p such that $\lambda = p(N-1)$ remains approximately constant:

1. compute the exact binomial degree distribution,
2. compute the Poisson approximation,
3. plot both on the same axes,
4. discuss how the agreement changes as N increases.

2.13.3.3 Exercise 19. Comparing $G(N, L)$ and $G(N, p)$ numerically

Choose values of N, L, and p such that

$$L \approx p\binom{N}{2}.$$

Then:

1. generate one graph from $G(N, L)$,
2. generate one graph from $G(N, p)$,
3. compare their edge counts, average degree, and clustering,
4. repeat this many times and summarize the differences.

2.13.3.4 Exercise 20. Empirical verification of expected average degree

For several values of N and p, generate many graphs from $G(N, p)$ and compute the empirical mean of the average degree.

Then compare it with the theoretical value

$$p(N-1).$$

Discuss the effect of the number of repetitions on the accuracy of the estimate.

2.13.3.5 Exercise 21. Empirical verification of expected clustering

For several values of N and p, generate many graphs from $G(N, p)$ and compute the empirical mean of the average clustering coefficient.

Then compare it with the theoretical value

$$p.$$

Explain why small discrepancies may remain in finite simulations.

2.13.3.6 Exercise 22. Degree table and empirical distribution from one graph

Generate one graph from $G(N, p)$ and produce:

1. a table of vertex degrees,
2. a table of degree counts,
3. the empirical degree distribution,
4. a plot of the empirical distribution.

Then compare the empirical plot with the theoretical binomial curve.

2.13.3.7 Exercise 23. Sensitivity to parameter changes

Fix N and vary p over a grid of values.

For each value of p, compute:

- average degree,
- empirical clustering,
- largest observed degree.

Plot each quantity against p and discuss the trends.

2.13.3.8 Exercise 24. Simulation report

Prepare a short report of approximately two pages containing:

- one derivation from the theory part of the chapter,
- at least two simulation figures,
- one comparison of empirical and theoretical degree distributions,
- one discussion of average degree and clustering in $G(N, p)$.

2.14 Suggested mini-project

A strong mini-project for this chapter is the following:

> Compare the theoretical and empirical behavior of the Erdős–Rényi model under several parameter regimes.

A complete submission should include:

- mathematical derivations,
- simulation code,
- figures,
- interpretation of the results,
- a short concluding discussion.

2.15 Concluding remarks

The Erdős–Rényi model is not meant to reproduce every feature of real networks. Its value lies in its clarity.

It gives a first probabilistic language for talking about networks:

- a graph can be generated by a random rule
- many graph properties can be studied through expectations
- independent edge formation leads naturally to binomial laws
- sparse large graphs often produce Poisson-like degree distributions
- independent wiring tends to create low clustering

That is why Erdős–Rényi graphs remain such an important first benchmark in network science.

Chapter 3

The Evolution of Random Graphs

3.1 Introduction

The previous chapter focused on the static properties of random graphs: the Erdős–Rényi models, average degree, degree distribution, and clustering coefficient. Those quantities describe the graph at a fixed stage.

A second and equally important question is how a random graph changes as links are gradually introduced. At the beginning there may be no links at all, so every vertex is isolated. When a small number of links is added, tiny connected groups appear. As the density of links increases, these groups merge into larger connected structures. Eventually one component becomes comparable in size to the full graph. This object is called the giant component.

The central theme of this chapter is that the emergence of large-scale connectivity is governed by a threshold phenomenon. In the Erdős–Rényi model, the expected degree

$$\langle k \rangle = p(N-1)$$

is the key parameter controlling that transition.

3.2 Learning goals

By the end of this chapter, it should be possible to:

- define connected graphs and connected components precisely
- define the size of a component and the notion of a giant component
- explain how a random graph evolves as the edge probability increases
- distinguish the subcritical, critical, supercritical, and connected regimes
- explain why the threshold $\langle k \rangle = 1$ is important
- explain why the connectedness threshold occurs near $\langle k \rangle = \ln N$
- verify these ideas numerically in `R`

3.3 Connectedness and components

3.3.1 Connected graph

Definition 3.1. Let $G = (V, E)$ be an undirected graph. The graph G is called **connected** if for every pair of distinct vertices $u, v \in V$, there exists a path in G from u to v.

This definition means that the graph forms one reachable structure. Starting from any vertex, it is possible to move to any other vertex by following a sequence of existing edges.

Connectedness is therefore a global property. It is not determined by one degree value or by one local motif. It asks whether the graph, as a whole, holds together.

If this condition fails, then the graph is disconnected. In that case the graph splits into separate connected pieces.

3.3.2 Connected component

Definition 3.2. Let $G = (V, E)$ be an undirected graph. A subgraph H of G is called a **connected component** of G if:

1. H is connected, and
2. H is maximal with respect to inclusion among connected subgraphs of G.

The word *connected* means that every pair of vertices in H can be joined by a path entirely contained in H.

The word *maximal* is equally important. It means that H cannot be enlarged by adding another vertex of G while keeping the subgraph connected. So a connected component is not merely a connected part of the graph. It is a connected part that is already as large as possible.

This gives a natural decomposition of a disconnected graph: the graph is partitioned into its connected components.

Remark 3.1. The connected components of a graph form a partition of the vertex set V. Every vertex belongs to exactly one connected component.

3.3.3 Size of a component

Definition 3.3. Let H be a connected component of a graph $G = (V, E)$. The **size** of H is the number of vertices in H, namely

$$|V(H)|.$$

This quantity is simple, but it becomes crucial in random graph theory. As a random graph evolves, the central question is how the component-size pattern changes. Do only small components exist, or does one component become comparable to the full graph?

3.3.4 Isolated vertex

Definition 3.4. A vertex $v \in V$ is called **isolated** if its degree is zero.

An isolated vertex is a connected component of size 1. So isolated vertices are not exceptions to the theory of components. They are simply the smallest possible components.

3.3.5 Example: component structure

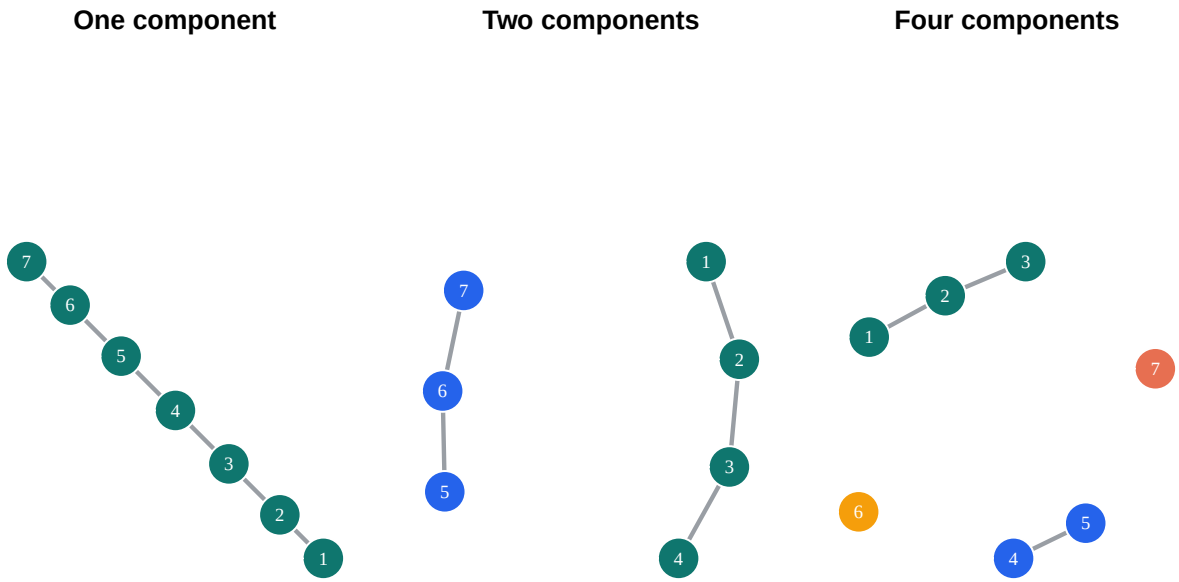


Figure 3.1: Three graphs with different component structure. Vertices in the same connected component share the same color.

The three graphs above illustrate the same idea in progressively more disconnected form:

- in the first graph, every vertex belongs to one single connected component
- in the second graph, the graph splits into two components
- in the third graph, the graph decomposes into several small pieces, including isolated vertices

3.3.6 Component-size table

Graph	Component sizes
One component	7
Two components	$4, 3$
Four components	$3, 2, 1, 1$

This table translates the picture into numerical structure and prepares the transition to the notion of a giant component.

3.4 Giant component

3.4.1 Definition

Definition 3.5. Let G_N be a graph on N vertices, and let $C_{\max}(G_N)$ denote its largest connected component. A sequence of random graphs is said to contain a **giant component** if

$$\frac{|C_{\max}(G_N)|}{N}$$

converges to a positive quantity as $N \to \infty$.

This definition should be interpreted carefully. Every finite graph has a largest connected component, but not every largest connected component is a giant component in the asymptotic sense.

The term *giant* means that the component contains a nonvanishing fraction of all vertices. It is therefore a macroscopic object, not merely the largest among several still-small pieces.

3.4.2 Why the giant component matters

The giant component marks the first stage at which the graph begins to behave like a true network at the global level.

Before it appears:

- most vertices belong only to tiny disconnected groups
- reachability is local
- no global connected backbone exists

After it appears:

- a positive fraction of vertices become mutually reachable
- one connected structure dominates the graph
- system-wide connectivity becomes possible

So the emergence of the giant component is the key event in the evolution of a random graph.

3.5 The evolution of a random graph

Consider the Erdős–Rényi model $G(N, p)$ with fixed N and variable parameter $p \in [0, 1]$.

At one extreme,

$$p = 0,$$

no edges are present, so every vertex is isolated.

At the other extreme,

$$p = 1,$$

every possible edge is present, so the graph is complete and hence connected.

The mathematically interesting behavior lies between these extremes. As p increases,

1. isolated vertices begin to join into tiny components,
2. small components merge into larger components,
3. one component begins to dominate,
4. eventually the graph becomes connected.

So the evolution of the graph is most naturally described through the evolution of its component structure.

3.5.1 Evolution through snapshots

To visualize this process clearly, the same vertex set and the same vertex positions are used in all snapshots below. The edge set evolves monotonically: once an edge appears, it remains present at later stages. This makes the sequence a true network evolution rather than a collection of unrelated random realizations.

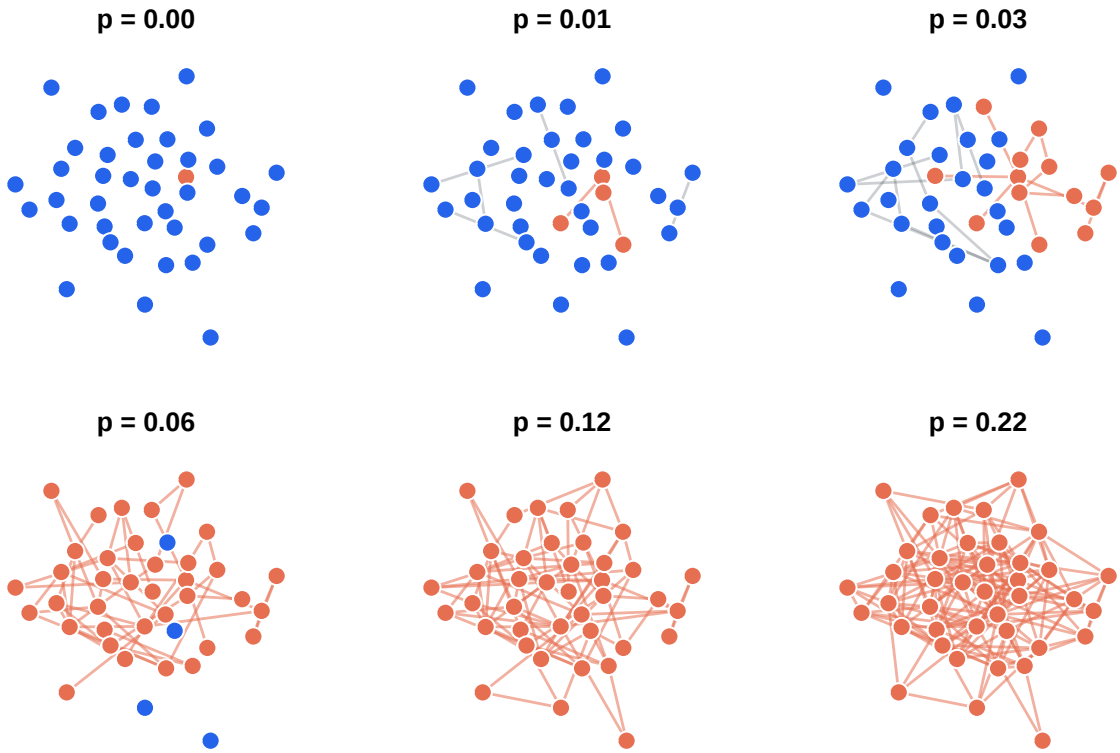


Figure 3.2: Evolution of a random graph on a fixed vertex set. Vertex positions are fixed across all snapshots, and the edge set grows monotonically with p. The largest connected component is highlighted.

These snapshots show several distinct structural stages.

- At $p = 0$, all vertices are isolated.
- For very small p, tiny components begin to appear.
- At intermediate values of p, components merge and one of them starts to dominate.
- At larger values of p, most vertices belong to a single large connected structure.

The qualitative transition is therefore from fragmentation to macroscopic connectivity.

3.6 Regimes of random-graph evolution

A convenient parameter for describing these regimes is the expected degree

$$\langle k \rangle = p(N-1).$$

This quantity is more informative than p alone because it measures the expected number of neighbors per vertex.

3.6.1 Subcritical regime

Definition 3.6. The random graph is in the **subcritical regime** when

$$\langle k \rangle < 1.$$

In this regime, no giant component exists. Almost all components are small, and isolated vertices are common. The graph contains connectivity, but only at a local scale.

3.6.2 Critical regime

Definition 3.7. The random graph is in the **critical regime** when

$$\langle k \rangle \approx 1.$$

This is the transition region. It is the stage at which the graph changes from a collection of small components into a graph containing a giant component.

In finite networks, the transition is not perfectly sharp. But as the system becomes large, the threshold becomes more pronounced.

3.6.3 Supercritical regime

Definition 3.8. The random graph is in the **supercritical regime** when

$$\langle k \rangle > 1.$$

This is the first regime in which a giant component exists. One connected component contains a positive fraction of all vertices, while many smaller components continue to coexist with it.

This regime is especially important because it is the first stage at which the graph has a genuine large-scale network structure.

3.6.4 Connected regime

Definition 3.9. The random graph enters the **connected regime** when the graph is connected with high probability as N becomes large. In the Erdős–Rényi model this occurs near

$$\langle k \rangle \approx \ln N,$$

or equivalently,

$$p \approx \frac{\ln N}{N}.$$

This threshold is much larger than the giant-component threshold $\langle k \rangle = 1$. So the emergence of a giant component and the graph becoming connected are two different events.

There is therefore a broad intermediate range in which the graph already has a giant component but is not yet fully connected.

3.7 Mathematical description of the giant component

A central quantity is the relative size of the giant component,

$$S = \frac{N_G}{N},$$

where N_G denotes the number of vertices in the largest connected component.

In the Erdős–Rényi model, the asymptotic fraction S of vertices contained in the giant component satisfies the self-consistency equation

$$S = 1 - e^{-\langle k \rangle S}.$$

This equation is a concise mathematical description of the giant component.

3.7.1 Interpretation of the self-consistency equation

The quantity $1 - S$ is the fraction of vertices that do **not** belong to the giant component.

A vertex fails to belong to the giant component only if all of its potential connections lead to vertices that also fail to belong to the giant component. In the Erdős–Rényi setting, that logic leads to the exponential form above.

So the equation

$$S = 1 - e^{-\langle k \rangle S}$$

expresses the self-consistent probability that a vertex belongs to the giant structure.

3.7.2 Near-threshold behavior

When $\langle k \rangle$ is only slightly larger than 1, the giant component is still small relative to its later size, but it is already macroscopic. In that near-threshold region,

$$S \sim 2(\langle k \rangle - 1) \qquad \text{as } \langle k \rangle \downarrow 1^+.$$

Equivalently,

$$N_G \sim 2(\langle k \rangle - 1)N.$$

So near the transition, the giant component grows linearly with the distance from the threshold.

3.7.3 Structure in the supercritical regime

In the supercritical regime, the graph has two coexisting structural parts:

1. one giant component containing a positive fraction of all vertices,
2. many finite components outside the giant component.

Near the threshold, those finite components are typically small and predominantly tree-like. By contrast, the giant component contains many cycles and loops. So the onset of the giant component is also the onset of genuinely complex global connectivity.

3.7.4 Connectedness threshold

The giant component threshold occurs near

$$\langle k \rangle = 1,$$

but the graph becomes connected only much later, near

$$\langle k \rangle = \ln N.$$

This is a major conceptual point. The existence of a giant component does **not** imply that the graph is connected.

So there is an intermediate range,

$$1 < \langle k \rangle < \ln N,$$

in which

- the graph has a giant component,
- many smaller components still exist,
- isolated vertices may still be present,
- the full graph is not yet connected.

Only when the average degree reaches order $\ln N$ do these remaining disconnected pieces disappear with high probability.

3.8 Numerical exploration

The structural transition can be observed numerically by generating many graphs for different values of p and tracking several global quantities.

```r
set.seed(2026)

n <- 500
k_grid <- seq(0.1, 8.0, length.out = 45)
p_grid <- k_grid / (n - 1)
reps <- 80

evolution_tbl <- map_dfr(p_grid, function(p) {
  map_dfr(seq_len(reps), function(rep_id) {
    g <- sample_gnp(n = n, p = p, directed = FALSE, loops = FALSE)
    comp <- components(g)$csize

    tibble(
      p = p,
      mean_degree = p * (n - 1),
      rep = rep_id,
      largest_component_fraction = max(comp) / n,
      number_of_components = length(comp),
      isolated_fraction = sum(comp == 1) / n,
      connected_indicator = as.integer(length(comp) == 1)
    )
  })
```

```r
})

evolution_summary <- evolution_tbl |>
  group_by(p, mean_degree) |>
  summarise(
    largest_component_fraction = mean(largest_component_fraction),
    number_of_components = mean(number_of_components),
    isolated_fraction = mean(isolated_fraction),
    connected_probability = mean(connected_indicator),
    connected_probability_se = sqrt(connected_probability * (1 - connected_probability) / n()),
    .groups = "drop"
  )

iso_fit <- isoreg(evolution_summary$mean_degree, evolution_summary$connected_probability)
evolution_summary$connected_probability_iso <- iso_fit$yf
```

3.8.1 Largest component fraction

```r
ggplot(evolution_summary, aes(x = mean_degree, y = largest_component_fraction)) +
  geom_line(color = course_cols$teal, linewidth = 1.2) +
  geom_point(color = course_cols$gold, size = 2.1) +
  geom_vline(xintercept = 1, color = course_cols$coral, linewidth = 1.0, linetype = 2) +
  geom_vline(xintercept = log(n), color = course_cols$blue, linewidth = 1.0, linetype = 3) +
  labs(
    title = "Emergence of the giant component",
    subtitle = "Red dashed line: giant-component threshold. Blue dotted line: connectedness threshold."
    x = "Expected degree E[k]",
    y = "Largest component fraction"
  )
```

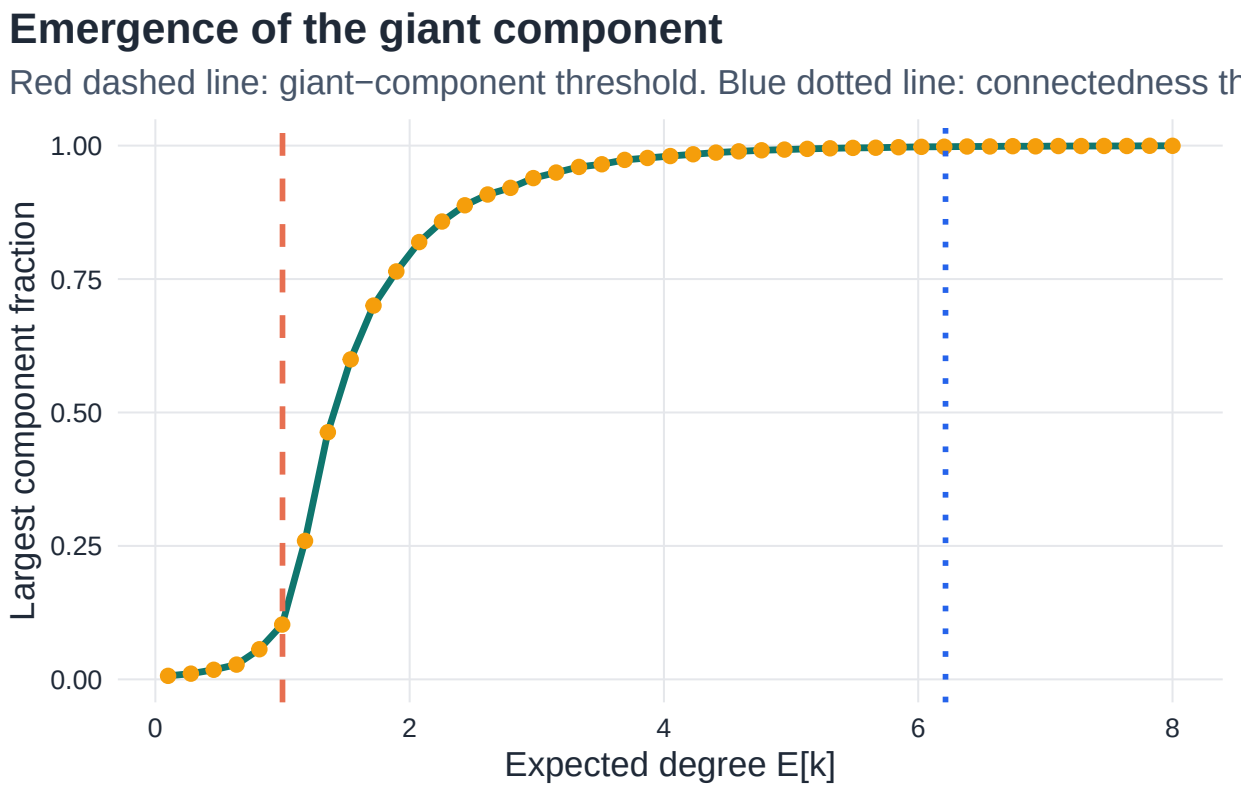


Figure 3.3: Average fraction of vertices in the largest connected component as a function of expected degree. The dashed vertical lines indicate the giant-component threshold and the connectedness threshold.

This plot captures the main phase transition. For small expected degree, the largest component occupies only a small fraction of the graph. Near $\langle k \rangle = 1$, the largest component begins to grow rapidly. Well before the graph becomes connected, one component already dominates the structure.

3.8.2 Number of connected components

```
ggplot(evolution_summary, aes(x = mean_degree, y = number_of_components)) +
  geom_line(color = course_cols$blue, linewidth = 1.2) +
  geom_point(color = course_cols$gold, size = 2.1) +
  geom_vline(xintercept = 1, color = course_cols$coral, linewidth = 1.0, linetype = 2) +
  geom_vline(xintercept = log(n), color = course_cols$blue, linewidth = 1.0, linetype = 3) +
  labs(
    title = "Decrease in the number of connected components",
    subtitle = "Disconnected pieces merge as the graph evolves",
    x = "Expected degree E[k]",
    y = "Average number of components"
  )
```

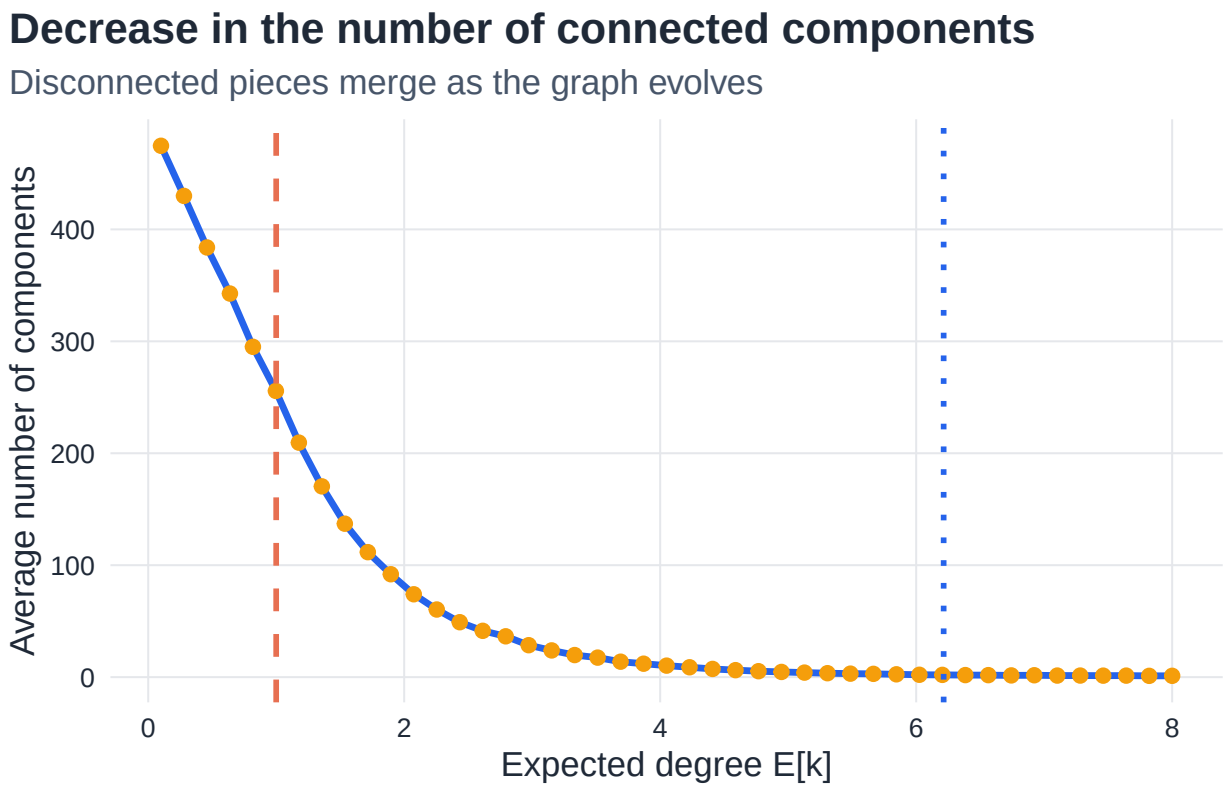


Figure 3.4: Average number of connected components as a function of expected degree.

This second plot reveals the same transition from another viewpoint. At small expected degree, the graph contains many disconnected pieces. As more links are added, these components merge, and the total number of components decreases.

3.8.3 Fraction of isolated vertices

```
ggplot(evolution_summary, aes(x = mean_degree, y = isolated_fraction)) +
  geom_line(color = course_cols$teal, linewidth = 1.2) +
  geom_point(color = course_cols$gold, size = 2.1) +
  geom_vline(xintercept = 1, color = course_cols$coral, linewidth = 1.0, linetype = 2) +
  geom_vline(xintercept = log(n), color = course_cols$blue, linewidth = 1.0, linetype = 3) +
  labs(
    title = "Disappearance of isolated vertices",
    subtitle = "Isolated vertices become rare as the network densifies",
    x = "Expected degree E[k]",
    y = "Fraction of isolated vertices"
  )
```

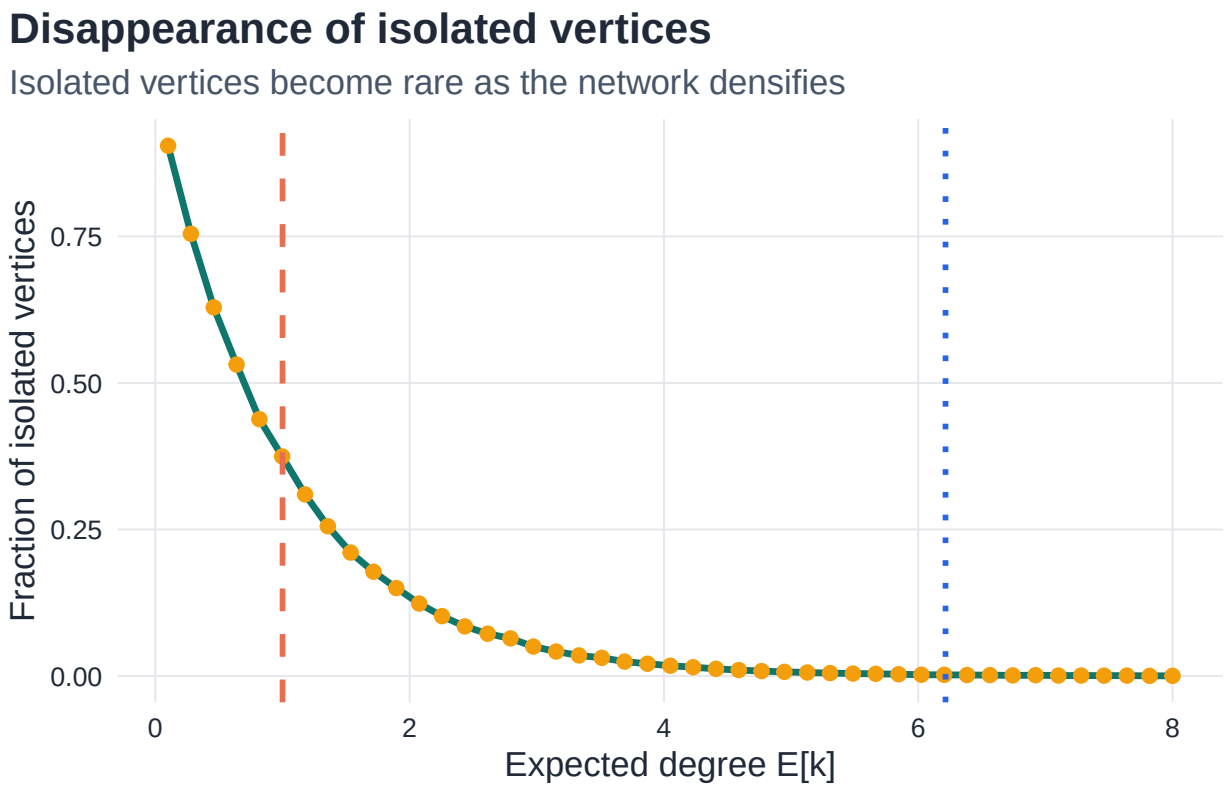


Figure 3.5: Average fraction of isolated vertices as a function of expected degree.

Isolated vertices are abundant at very small edge density, but they disappear as the graph evolves. Their disappearance is one of the main signals that the graph is approaching connectedness.

3.8.4 Probability of connectedness

```
ggplot(evolution_summary, aes(x = mean_degree)) +
  geom_ribbon(
    aes(
      ymin = pmax(connected_probability - 1.96 * connected_probability_se, 0),
      ymax = pmin(connected_probability + 1.96 * connected_probability_se, 1)
    ),
    fill = course_cols$blue,
    alpha = 0.12
  ) +
  geom_point(aes(y = connected_probability), color = course_cols$gold, size = 1.9) +
  geom_line(aes(y = connected_probability_iso), color = course_cols$coral, linewidth = 1.2) +
  geom_vline(xintercept = 1, color = course_cols$coral, linewidth = 1.0, linetype = 2) +
  geom_vline(xintercept = log(n), color = course_cols$blue, linewidth = 1.0, linetype = 3) +
  labs(
    title = "Connectedness probability",
    subtitle = "The true probability is monotone increasing; the fitted line enforces that theoretical p
    x = "Expected degree E[k]",
    y = "Probability that the graph is connected"
  )
```

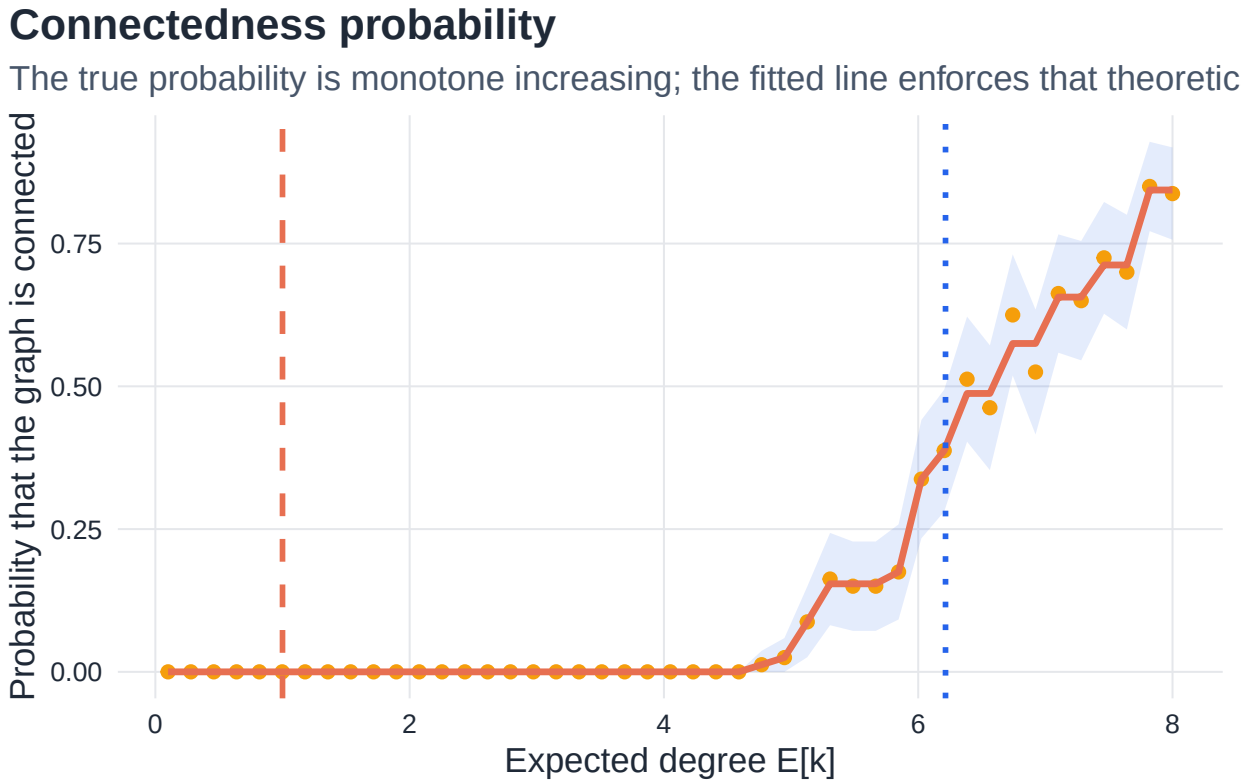


Figure 3.6: Estimated connectedness probability as a function of expected degree. Points show Monte Carlo estimates; the solid curve is the monotone isotonic fit.

For fixed N, the true connectedness probability is a monotone increasing function of p. Small downward wiggles in raw simulation estimates are caused by Monte Carlo noise. The points in the figure are the empirical estimates from a finite number of repetitions, while the solid line is an isotonic fit, which enforces the correct monotone behavior.

This last plot makes the distinction between the two thresholds especially clear. The giant component appears near $\langle k \rangle = 1$, but the probability that the graph is fully connected remains small until the expected degree is of order $\ln N$.

3.9 Summary of regimes

Regime	Condition	Main structural property
Subcritical	$\langle k \rangle < 1$	Only small connected components are present
Critical	$\langle k \rangle \approx 1$	The giant component begins to emerge
Supercritical	$\langle k \rangle > 1$	A giant component coexists with many finite components
Connected	$\langle k \rangle \approx \ln N$	The full graph is connected with high probability

This table summarizes the evolution of the Erdős–Rényi random graph in terms of its component structure.

3.10 In-class questions

1. Why is connectedness a global property of a graph?

2. What role does maximality play in the definition of a connected component?
3. Why is an isolated vertex still a connected component?
4. What distinguishes a largest component from a giant component?
5. Why does the threshold $\langle k \rangle = 1$ matter?
6. Why does the connectedness threshold occur later, near $\langle k \rangle = \ln N$?
7. Why can a graph have a giant component and still be disconnected?

3.11 End-of-chapter exercises

The exercises below are grouped into three categories:

1. mathematical exercises,
2. conceptual and analytical exercises,
3. simulation and coding exercises.

They are designed to move from formal understanding to computational verification.

3.11.1 Part I. Mathematical exercises

3.11.1.1 Exercise 1. Components and partitions

Let $G = (V, E)$ be an undirected graph.

1. Prove that every vertex of G belongs to at least one connected component.
2. Prove that no vertex can belong to two distinct connected components.
3. Deduce that the connected components of G form a partition of the vertex set V.

3.11.1.2 Exercise 2. Isolated vertices

Let $v \in V$ be a vertex with degree zero.

1. Prove that $\{v\}$ is a connected subgraph.
2. Prove that $\{v\}$ is maximal among connected subgraphs containing v.
3. Conclude that an isolated vertex is a connected component of size 1.

3.11.1.3 Exercise 3. Giant component criterion

Let $C_{\max}(G_N)$ denote the largest connected component of a graph on N vertices.

1. Explain why every finite graph has a largest connected component.
2. Give an example of a sequence of graphs for which
$$\frac{|C_{\max}(G_N)|}{N} \to 0.$$
3. Give an example of a sequence of graphs for which
$$\frac{|C_{\max}(G_N)|}{N} \to c > 0.$$
4. Explain why the second case corresponds to the existence of a giant component.

3.11.1.4 Exercise 4. Threshold written in terms of p

In the Erdős–Rényi model $G(N, p)$, the expected degree is

$$\langle k \rangle = p(N-1).$$

1. Show that the condition
$$\langle k \rangle = 1$$
implies
$$p_c = \frac{1}{N-1}.$$
2. Show that for large N this becomes
$$p_c \sim \frac{1}{N}.$$
3. Explain why this implies that the critical edge density becomes very small in large networks.

3.11.1.5 Exercise 5. Connectedness threshold

The connectedness threshold in the Erdős–Rényi model occurs near

$$\langle k \rangle \approx \ln N.$$

1. Show that this corresponds to
$$p \approx \frac{\ln N}{N-1}.$$
2. Compare this threshold with the giant-component threshold
$$p_c \approx \frac{1}{N-1}.$$
3. Prove that
$$\ln N > 1 \qquad \text{for all } N > e,$$
and explain why connectedness must occur later than the emergence of the giant component.

3.11.1.6 Exercise 6. The self-consistency equation

The asymptotic fraction S of nodes in the giant component satisfies

$$S = 1 - e^{-\langle k \rangle S}.$$

1. Verify that $S = 0$ is always a solution.
2. Show that if $\langle k \rangle < 1$, then the tangent of the right-hand side at $S = 0$ has slope smaller than 1.
3. Explain why this indicates that no positive solution exists in the subcritical regime.
4. Show that when $\langle k \rangle > 1$, the slope at $S = 0$ exceeds 1.
5. Explain why this allows the existence of a positive solution corresponding to a giant component.

3.11.1.7 Exercise 7. Near-threshold expansion

Starting from

$$S = 1 - e^{-\langle k \rangle S},$$

assume that S is small and that $\langle k \rangle = 1 + \varepsilon$ with $\varepsilon > 0$ small.

1. Expand the exponential up to second order in S.
2. Show that, to leading order,
$$S \approx 2(\langle k \rangle - 1).$$
3. Interpret this result in terms of the growth of the giant component just above the threshold.

3.11.2 Part II. Conceptual and analytical exercises

3.11.2.1 Exercise 8. Regime classification

For each of the following values of $\langle k \rangle$, determine whether the graph is expected to be in the subcritical, critical, supercritical, or connected regime:

1. $\langle k \rangle = 0.4$
2. $\langle k \rangle = 0.95$
3. $\langle k \rangle = 1.8$
4. $\langle k \rangle = \ln 500$

In each case, describe the expected component structure.

3.11.2.2 Exercise 9. Giant component versus connected graph

Explain carefully why the following two statements are not equivalent:

1. the graph has a giant component,
2. the graph is connected.

Your answer should include:

- a formal distinction,
- an explanation in terms of component sizes,
- an example of a graph that has a giant component but is not connected.

3.11.2.3 Exercise 10. Structural interpretation of the regimes

Write a short mathematical discussion comparing the four regimes:

- subcritical,
- critical,
- supercritical,
- connected.

For each regime, describe:

- the typical component sizes,
- whether isolated vertices are common,
- whether a giant component exists,
- whether the graph is connected.

3.11.2.4 Exercise 11. Why the emergence is a phase transition

The emergence of the giant component is often described as a phase transition.

Write a short explanation of this statement. Your answer should address:

- what quantity changes qualitatively,
- what the threshold parameter is,
- why the transition is not merely a gradual visual change,
- why this phenomenon is global rather than local.

3.11.2.5 Exercise 12. Trees and cycles outside the giant component

In the supercritical regime, many finite components outside the giant component are tree-like.

1. Explain why a finite tree on m vertices has exactly $m - 1$ edges.
2. Explain why the appearance of cycles indicates additional structural complexity.
3. Discuss why the giant component is expected to contain cycles, whereas many small finite components remain tree-like near the threshold.

3.11.3 Part III. Simulation and coding exercises

3.11.3.1 Exercise 13. Component-size computation

Write an `R` function that takes an undirected graph `g` and returns:

- the number of connected components,
- the vector of component sizes,
- the size of the largest connected component,
- the fraction of vertices in the largest connected component.

Test the function on at least three hand-constructed graphs.

3.11.3.2 Exercise 14. Evolution of the largest component

For fixed $N = 500$, simulate the model $G(N, p)$ over a grid of p values corresponding to

$$0.1 \leq \langle k \rangle \leq 8.$$

For each value of p, estimate the average largest-component fraction over repeated realizations.

Tasks:

1. plot the largest-component fraction against $\langle k \rangle$,
2. indicate the line $\langle k \rangle = 1$,
3. explain where the giant component begins to emerge.

3.11.3.3 Exercise 15. Number of components and isolated vertices

Using the same simulations as in Exercise 14, compute and plot:

1. the average number of connected components,
2. the average fraction of isolated vertices.

Discuss how these two quantities evolve as $\langle k \rangle$ increases.

3.11.3.4 Exercise 16. Empirical connectedness probability

For fixed $N = 500$, estimate the probability that $G(N, p)$ is connected for a grid of values of p.

Tasks:

1. plot the empirical connectedness probability against $\langle k \rangle$,
2. indicate the line $\langle k \rangle = \ln N$,
3. explain why the curve should be monotone increasing in theory,
4. explain why Monte Carlo estimates may show small fluctuations.

3.11.3.5 Exercise 17. Finite-size comparison

Repeat the largest-component simulation for

$$N = 100, \quad 250, \quad 500, \quad 1000.$$

For each value of N:

1. plot the largest-component fraction against $\langle k \rangle$,
2. compare the sharpness of the transition near $\langle k \rangle = 1$,
3. discuss how the finite-size behavior changes as N increases.

3.11.3.6 Exercise 18. Connectedness threshold and system size

For each of the values

$$N = 100, \quad 250, \quad 500, \quad 1000,$$

estimate numerically the value of $\langle k \rangle$ at which the graph becomes connected with probability approximately 0.5.

Then compare your estimates with $\ln N$.

Discuss whether the numerical behavior supports the theoretical scaling of the connectedness threshold.

3.11.3.7 Exercise 19. Monotone network evolution

Construct a sequence of graphs on a fixed set of vertices such that the edge set grows monotonically as p increases.

Tasks:

1. keep vertex positions fixed across all snapshots,

2. display at least six stages of evolution,
3. highlight the largest component at each stage,
4. explain how the component structure changes visually.

3.11.3.8 Exercise 20. Numerical solution of the giant-component equation

Consider the equation

$$S = 1 - e^{-\langle k \rangle S}.$$

1. Write an `R` function that solves this equation numerically for a given value of $\langle k \rangle$.
2. Compute the solution for a grid of values of $\langle k \rangle$ between 0 and 5.
3. Plot the resulting theoretical curve.
4. Compare it with your simulation-based estimate of the largest-component fraction.

3.11.3.9 Exercise 21. Short computational report

Prepare a short report of approximately two pages based on your simulations. The report should include:

- a brief statement of the model,
- at least two figures,
- one discussion of the threshold near $\langle k \rangle = 1$,
- one discussion of the connectedness threshold near $\langle k \rangle = \ln N$,
- a concluding paragraph interpreting the emergence of global connectivity.

3.12 Suggested project-style extension

A strong mini-project for this chapter is the following:

> Compare the simulated evolution of $G(N, p)$ with the theoretical equations for giant-component emergence and connectedness threshold.

A complete submission should contain:

- mathematical derivations,
- simulation code,
- plots,
- interpretation of the results.

3.13 Concluding remarks

The evolution of a random graph is one of the clearest demonstrations of how global structure can arise from simple local randomness.

At first the graph contains only isolated vertices and very small components. As edges are added, these small pieces merge. Near the threshold $\langle k \rangle = 1$, a giant component emerges. Much later, near $\langle k \rangle = \ln N$, the graph becomes connected with high probability.

So the emergence of a network is not a smooth undifferentiated process. It is organized by precise structural thresholds, and those thresholds can be described mathematically in terms of component size and expected degree.